SLEB Smart Hub

Building Energy Efficiency Assessment (BEEA) & Industrial Energy Efficiency Assessment (IEEA) API Documentation

Ver 1.0

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1. Security and Data Integrity

1.1. Interface Access Method

- HTTPs based

1.2. Authorization

The token must be included in all requests' HTTPs Header with key "Authorization" and value format Bearer <token>

1.3. Message Specification

- Build APIs in JSON.
- Input parameters are case sensitive.

2. Interface Description

2.1. Authentication

- Authentication with user name and password.
- Return API access token.

URL: https://sleb.sg/slebapi/beea/login
Method: POST

Request

Name	Type	Required	Description
username	string	Yes	*required* User name
password	string	Yes	*required* Password

Response

Name	Type	Description
errcode	int	Error code for the API (0: ok, 1: error)
errmsg	string	Message
token	string	Access token, only if login successful

Examples

Example Request

```
{
    “username”: “name”,
    “password”: “password”
}
```

Example Response

```
{
  “result”: {
            “errcode”:  0,
            “errmsg”: “Login Success”
  },
  “data”: {
            “token”: “0123456789abcdefghijklmnopqrstuvwxyz”
  }
}
```

2.2. Generate Building/Industrial Facility Energy Efficiency Assessment Report

2.2.1 New Development Project for Building

- Generate a building energy efficiency assessment report of a new development project.
- Return API response

URL: https://sleb.sg/slebapi/beea/project/report/newdev
Method: POST

Request

Category	Data Point	Variable Name	Type	Description
-	Project Name	ProjectName	string	*required* Project name
-	Project Address	ProjectAddress	string	*required* Project address
-	Project Postal Code	ProjectPostalCode	string	*required* Project postal code
-	Project Country	ProjectCountry	string	*required* Project country
-	Project Gross Floor Area	ProjectGFA	string	*required* Project gross floor area (excluding car park area) in m^2 input range: (0, ∞)
-	Project Status	ProjectStatus	string	*required* Project status input range: {"New Development}
-	Building Data	BuildingData	object	*required* Building data
-	Building Information	BuildingInfo	object	*required* Building data under category of building information
-	Facade	Facade	object	*required* Building data under category of facade
-	Air Conditioning System	ACSystem	object	*required* Building data under category of air conditioning (A/C) system
-	Air Distribution System	ADSystem	object	*required* Building data under category of air distribution (A/D) system
-	Lighting System	LightingSystem	object	*required* Building data under category of lighting system
-	Mechanical Ventilation System	MVSystem	object	*required* Building data under category of mechanical ventilation (MV) system
-	Other System	OtherSystem	object	*required* Building data under category of other system
Building Information	Building Name	BuildingName	string	*required* Building Name input range: free text
	Building Gross Floor Area	GFA	string	*required* Building Gross Floor Area (excluding car park area) in m^2 input range: (0, ∞)
	Building Type	BuildingType	string	*required* Building Type

				input range: {“Office Building”, “Retail Building”, “Hotel”, “Healthcare Facility”, “Educational Institution”, “Industrial Building”, “Residential Building”, “Mixed Development”, “Others”}
	Number of Storeys	Storey	int	*required* Number of Storeys input range: 1, 300
	Year of Temporary Occupation Permit or Certificate of Statutory Completion	YearTOP	string	*required* Year of Temporary Occupation Permit or Certificate of Statutory Completion input range: 1900-9999
	Food and Beverage Area	FBArea	string	*required* Food and Beverage Area in m^2 input range: (0, ∞)
	Supermarket Area	SupermarArea	string	*required* Supermarket Area in m^2 input range: (0, ∞)
	Data Centre/Server Room Area	DCArea	string	*required* Data Centre/Server Room Area in m^2 input range: (0, ∞)
	Total Building Operation	Schedule	string	*required* Total Building Operation in hours per week input range: (0, 168)
	Percentage of Air-Conditioned Area	PACArea	string	*required* Percentage of Air-Conditioned Area in % input range: (0, 100)
	Air Conditioning System Type	ACType	string	*required* Air Conditioning System Type input range: {“Water-cooled Chilled Water Plant”, “Air-cooled Chilled Water Plant”, “Unitary Air-conditioning System”, “District Cooling System”}
	Air Distribution System Type	ExistingADType	string	*required* Air Distribution System Type input range: {“AHU-CAV”, “AHU-VAV”, “FCU-CAV”, “FCU-VAV”}
	Total Mechanically Ventilated Area	MVArea	string	*required* Total Mechanically Ventilated Area in m^2 input range: (0, ∞)
Facade	Percentage of Natural	PNVArea	string	*required* Percentage of Natural Ventilation Area Converted from

	Ventilation Area Converted from AC Area			AC Area (Design of natural ventilation) in % input range: (0, PACArea)
	Reduced Window to Wall Ratio	WWR	string	*required* Reduced Window to Wall Ratio input range: (0, 0.4)
	Roof U-Value (Insulation of roof)	RoofU	string	*required* Roof U-Value (Insulation of roof) in W/m²·K input range: (0.2, 1.5)
	Wall U-Value	WallU	string	*required* Wall U-Value (Insulation of external walls) in W/m²·K input range: (0.2, 1.5)
	Glass U-Value	GlassU	string	*required* Glass U-Value (Use of high-performance glass) in W/m²·K input range: (0.5, 2.8)
	Glass Solar Heat Gain Coefficient	GlassSHGC	string	*required* Solar Heat Gain Coefficient (Use of high-performance glass) input range: (0.1, 0.35)
	Solar Window Film U-Value	FilmU	string	*required* Solar Window Film U-Value (Use of solar window film) in W/m²·K input range: (1, 10)
	Film Solar Heat Gain Coefficient	FilmSHGC	string	*required* Solar Heat Gain Coefficient (Use of solar window film) input range: (0.05, 0.6)
	Window Frame U-Value	FrameU	string	*required* Window Frame U-Value (Use of thermally broken window frames) in W/m²·K input range: (0.4, 1.2)
	Solar Reflectivity	PaintSR	string	*required* Solar Reflectivity (Use of reflective paint/cool paint) input range: (0.3, 0.9)
	Shading Coefficient	Shading	string	*required* Shading Coefficient (Installation of external shading devices) input range: (0.1, 0.9)
Air Conditioning (A/C) System	Energy-Efficient Chiller	ChillerEffi	string	*required* Energy-Efficient Chiller (For air conditioning system type of water-cooled chilled water plant, air-cooled chilled water plant) in kW/RT

				input range: 0.4, 0.9 (if ACType == “Water-cooled Chilled Water Plant”) 0.8, 1.5 (if ACType == “Air-cooled Chilled Water Plant”)
	Energy-Efficient Chilled Water Pump	CHWPEffi	string	*required* Energy-Efficient Chilled Water Pump (For air conditioning system type of water-cooled chilled water plant, air-cooled chilled water plant, district cooling system) in kW/m³/s input range: (160, 350)
	Energy Efficient Condensing Water Pump	CWPEffi	string	*required* Energy Efficient Condensing Water Pump (For air conditioning system type of water-cooled chilled water plant) in kW/m³/s input range: (140, 300)
	Energy-Efficient Cooling Tower	CTEffi	string	*required* Energy-Efficient Cooling Tower (For air conditioning system type of water-cooled chilled water plant) in L/s/kW input range: (1.7, 7)
	Air Conditioning System Optimisation	ACOpt	string	*required* Air Conditioning System Optimisation (For air conditioning system type of water-cooled chilled water plant, air-cooled chilled water plant) input range: {“Yes”}
	Energy-Efficient Unitary Air-Conditioning System	UnitaryEffi	string	*required* Energy-Efficient Unitary Air-Conditioning System (For air conditioning system type of unitary air-conditioning system) in kW/RT input range: (0.6, 1.1)
Air Distribution (A/D) System	Use of Variable Speed Drive (VSD) in AHU/FCU	ADType	string	*required* Use of Variable Speed Drive (VSD) in AHU/FCU (For air distribution system type of AHU-CAV and FCU-CAV) input range: {“AHU-VAV”, “FCU-VAV”}
	Percentage Use of Alternative Cooling Technologies	PACT	string	*required* Percentage Use of Alternative Cooling Technologies in % input range: (0, 100)

	Energy Efficient Air Distribution Fan	ADFanEffi	string	*required* Energy Efficient Air Distribution Fan (For air conditioning system type of water-cooled chilled water plant, air-cooled chilled water plant, district cooling system) in W/CMH input range: 0.05, 0.18 (ExistingADType ==FCU-CAV and ADType==None) 0.2, 0.5 (ExistingADType ==AHU-CAV and ADType==None) 0.3, 0.6 (ExistingADType ==AHU-VAV or ADType==AHU-VAV) 0.06, 0.22 (ExistingADType ==FCU-VAV or ADType==FCU-VAV)
	Demand-Controlled Ventilation of AC Area	DCV	string	*required* Demand-Controlled Ventilation of AC Area (For air conditioning system type of water-cooled chilled water plant, air-cooled chilled water plant, district cooling system) input range: {"Yes"}
	Use of Energy Recovery Ventilation System	ERVEffi	string	*required* Use of Energy Recovery Ventilation System (For air conditioning system type of water-cooled chilled water plant, air-cooled chilled water plant, district cooling system) in % input range: (30, 95)
	Percentage Use of Fans to Offset Cooling Load	PFan	string	*required* Percentage Use of Fans to Offset Cooling Load in % input range: (0, 100)
Lighting System	LED Efficacy	LEDEfficacy	string	*required* LED Efficacy (Use of energy efficient LED lights) in lm/W input range: (70, 200)
	Percentage of Use of Energy Efficient LED Lights	PLED	string	*required* Percentage of Use of Energy Efficient LED Lights in % input range: (0, 100)
	Smart Lighting Control Percentage of Application	PLigCrl	string	*required* Percentage of Application (Use of smart lighting control) in % input range: (0, 100)
	Daylighting Percentage of Application	PDaylight	string	*required* Percentage of Application (Use of daylighting) in % input range: (0, 100)

Mechanical Ventilation (MV) System	Energy-Efficient Mechanical Ventilation Fan	MVFanEffi	string	*required* Energy-Efficient Mechanical Ventilation Fan (Fan Efficiency) in W/CMH input range: (0.05, 0.3)
	Demand-Controlled Ventilation of MV Area	PMVCrl	string	*required* Demand-Controlled Ventilation of MV Area in % input range: (0, 100)
Other System	Use of Smart Plug Load Control	PLC	string	*required* Use of Smart Plug Load Control input range: {"Yes"}
	Use of Energy Efficient Appliances	EfficientApp	string	*required* Use of Energy Efficient Appliances (For building type of residential building) input range: {"Yes"}
	Use of Energy Efficient Hot Water System	HWCOP	string	*required* Use of Energy Efficient Hot Water System (For building type of hotel, healthcare facility, residential building) input range: (2, 5)
	Percentage Use of Solar Hot Water Collectors	PSolarHW	string	*required* Percentage Use of Solar Hot Water Collectors (for building type of hotel, healthcare facility, residential building) in % input range: (0, 100)
	Use of Energy Efficient Lift	EfficientLift	string	*required* Use of Energy Efficient Lift input range: {"Yes"}
	Use of Energy Efficient Escalator	EfficientEscalator	string	*required* Use of Energy Efficient Escalator input range: {"Yes"}
	Implementation of Building Energy Management System	BEMS	string	*required* Implementation of Building Energy Management System input range: {"Yes"}
	Rooftop Solar Panel Efficiency	PVEffi	string	*required* Rooftop Solar Panel Efficiency in % input range: (0, 100)
	Rooftop Solar Panel Area	PVArea	string	*required* Rooftop Solar Panel Area in m^2 input range: (0, ∞)
	BIPV Solar Panel Efficiency	BIPVEffi	string	*required* BIPV Solar Panel Efficiency in %

				input range: (0, 100)
	BIPV Solar Panel Area	BIPVArea	string	*required* BIPV Solar Panel Area in m^2 input range: (0, ∞)

Response

Variable Name	Type	Description
errcode	int	Error code for the API (0: ok, 1: error)
errmsg	string	Message
report	string	Generated building energy efficiency assessment report in Base64 format

Examples

Example Request


```
{
    “ProjectName”: “Project AAA”,
    “ProjectAddress”: “Project Address”,
    “ProjectPostalCode”: “123456”,
    “ProjectCountry”: “Project Country”,
    “ProjectGFA”: “10000”,
    “ProjectStatus”: “New Development”,
    “BuildingData”:
        {
            “BuildingInfo”:
                {
                    “BuildingName”: “Building AAA”,
                    “GFA”: “10000”,
                    “BuildingType”: “Office Building”,
                    “Storey”: 10,
                    “YearTOP”: “2025”,
                    “FBArea”: “0”,
                    “SupermarArea”: “0”,
                    “DCArea”: “100”,
                    “Schedule”: “55”,
                    “PACArea”: “80”,
                    “ACType”: “Water-cooled Chilled Water Plant”,
                    “ExistingADType”: “AHU-VAV”,
                    “MVArea”: “2000”
                }
            ,
            “Facade”:
                {
                    “PNVArea”: “10”,
                    “WWR”: “0.3”,
                    “RoofU”: “0.6”,
                    “WallU”: “0.6”,
                    “GlassU”: “1.9”,
                    “GlassSHGC”: “0.25”,
                    “FilmU”: “4”,
                    “FilmSHGC”: “0.3”,
```

```
                "FrameU": "0.7",
                "PaintSR": "0.6",
                "Shading": "0.6"
            }
        ,
        "ACSystem":
            {
                "ChillerEffi": "0.52",
                "CHWPEffi": "230",
                "CWPEffi": "210",
                "CTEffi": "5.2",
                "ACOpt": "Yes"
            }
        ,
        "ADSystem":
            {
                "PACT": "20",
                "ADFanEffi": "0.46",
                "DCV": "Yes",
                "ERVEffi": "70",
                "PFan": "50"
            }
        ,
        "LightingSystem":
            {
                "LEDEfficacy": "160",
                "PLED": "90",
                "PLigCrl": "50",
                "PDaylight": "20"
            }
        ,
        "MVSystem":
            {
                "MVFanEffi": "0.15",
                "PMVCrl": "50"
            }
        ,
        "OtherSystem"
            {
                "PLC": "Yes",
                "EfficientLift": "Yes",
                "EfficientEscalator": "Yes",
                "BEMS": "Yes",
                "PVEffi": "19",
                "PVArea": "100",
                "BIPVEffi": "19",
                "BIPVArea": "100"
            }

    }

}
```


Example Response

```
{
    “result”:
        {
            “errcode”: 0,
            “errmsg”: “null”
        },
    “data”:
        {
            “report”: “JVBERi0xLjMKMyAwIG9iago8PC9UeXBlIC9QYWdlCi9QYXJlbnQgMSAwIFIK...”
        }
}
```

2.2.2 Retrofitting Project for Building

- Generate a building energy efficiency assessment report of a retrofitting project.
- Return API response

URL: https://sleb.sg/slebapi/beea/project/report/retrofit
Method: POST

Request

Category	Data Point	Variable Name	Type	Description
-	Project Name	ProjectName	string	*required* Project name
-	Project Address	ProjectAddress	string	*required* Project address
-	Project Postal Code	ProjectPostalCode	string	*required* Project postal code
-	Project Country	ProjectCountry	string	*required* Project country
-	Project Gross Floor Area	ProjectGFA	string	*required* Project gross floor area (excluding car park area) in m^2 input range: (0, ∞)
-	Project Status	ProjectStatus	string	*required* Project status input range: {"Retrofitting Project"}
-	Building Data	BuildingData	object	*required* Building data
-	Building Information	BuildingInfo	object	*required* Building data under category of building information
-	Facade	Facade	object	*required* Building data under category of facade
-	Air Conditioning System	ACSystem	object	*required* Building data under category of air conditioning (A/C) system
-	Air Distribution System	ADSystem	object	*required* Building data under category of air distribution (A/D) system
-	Lighting System	LightingSystem	object	*required* Building data under category of lighting system
-	Mechanical Ventilation System	MVSystem	object	*required* Building data under category of mechanical ventilation (MV) system
-	Other System	OtherSystem	object	*required* Building data under category of other system
Building Information	Building Name	BuildingName	string	*required* Building Name input range: free text
	Building Gross Floor Area	GFA	string	*required* Building Gross Floor Area (excluding car park area) in m^2 input range: (0, ∞)
	Building Type	BuildingType	string	*required* Building Type input range: {"Office Building", "Retail Building", "Hotel",

				"Healthcare Facility", "Educational Institution", "Industrial Building", "Residential Building", "Mixed Development", "Others"}
	Number of Storeys	Storey	int	*required* Number of Storeys input range: 1, 300
	Year of Temporary Occupation Permit or Certificate of Statutory Completion	YearTOP	string	*required* Year of Temporary Occupation Permit or Certificate of Statutory Completion input range: 1900-9999
	Food and Beverage Area	FBArea	string	*required* Food and Beverage Area in m^2 input range: (0, ∞)
	Supermarket Area	SupermarArea	string	*required* Supermarket Area in m^2 input range: (0, ∞)
	Data Centre/Server Room Area	DCArea	string	*required* Data Centre/Server Room Area in m^2 input range: (0, ∞)
	Total Building Operation	Schedule	string	*required* Total Building Operation in hours per week input range: (0, 168
	Average Monthly Energy Consumption	MonthlyConsumption	string	*required* Average Monthly Energy Consumption in kWh input range: (0, ∞)
	Percentage of Air-Conditioned Area	PACArea	string	*required* Percentage of Air-Conditioned Area in % input range: 0, 100
	Air Conditioning System Type	ACType	string	*required* Air Conditioning System Type input range: {"Water-cooled Chilled Water Plant", "Air-cooled Chilled Water Plant", "Unitary Air-conditioning System", "District Cooling System"}
	Air Conditioning System Efficiency	ExistingACEffi	string	*required* Air Conditioning System Efficiency in kW/RT input range: (0, 3
	Air Distribution System Type	ExistingADType	string	*required* Air Distribution System Type input range: {"AHU-CAV", "AHU-VAV", "FCU-CAV", "FCU-VAV"}

	Air Distribution System Efficiency	ExistingADEffi	string	*required* Air Distribution System Efficiency in kW/RT input range: (0, 1
	Percentage Use of LED Lights	ExistingPLED	string	*required* Percentage Use of LED Lights in % input range: 0, 100
	Total Mechanically Ventilated Area	MVArea	string	*required* Total Mechanically Ventilated Area in m^2 input range: 0, ∞)
Facade	Percentage of Natural Ventilation Area Converted from AC Area	PNVArea	string	*required* Percentage of Natural Ventilation Area Converted from AC Area (Design of natural ventilation) in % input range: 0, PACArea
	Reduced Window to Wall Ratio	WWR	string	*required* Reduced Window to Wall Ratio input range: 0, 0.4
	Roof U-Value (Insulation of roof)	RoofU	string	*required* Roof U-Value (Insulation of roof) in $W/m^2 \cdot K$ input range: 0.2, 1.5
	Wall U-Value	WallU	string	*required* Wall U-Value (Insulation of external walls) in $W/m^2 \cdot K$ input range: 0.2, 1.5
	Glass U-Value	GlassU	string	*required* Glass U-Value (Use of high-performance glass) in $W/m^2 \cdot K$ input range: 0.5, 2.8
	Glass Solar Heat Gain Coefficient	GlassSHGC	string	*required* Solar Heat Gain Coefficient (Use of high-performance glass) input range: 0.1, 0.35
	Solar Window Film U-Value	FilmU	string	*required* Solar Window Film U-Value (Use of solar window film) in $W/m^2 \cdot K$ input range: 1, 10
	Film Solar Heat Gain Coefficient	FilmSHGC	string	*required* Solar Heat Gain Coefficient (Use of solar window film) input range: 0.05, 0.6
	Window Frame U-Value	FrameU	string	*required* Window Frame U-Value (Use of thermally broken window frames) in $W/m^2 \cdot K$ input range: 0.4, 1.2

	Solar Reflectivity	PaintSR	string	*required* Solar Reflectivity (Use of reflective paint/cool paint) input range: 0.3, 0.9
	Shading Coefficient	Shading	string	*required* Shading Coefficient (Installation of external shading devices) input range: 0.1, 0.9
Air Conditioning (A/C) System	Energy-Efficient Chiller	ChillerEffi	string	*required* Energy-Efficient Chiller (For air conditioning system type of water-cooled chilled water plant, air-cooled chilled water plant) in kW/RT input range: 0.4, 0.9 (if ACType == "Water-cooled Chilled Water Plant") 0.8, 1.5 (if ACType == "Air-cooled Chilled Water Plant")
	Energy-Efficient Chilled Water Pump	CHWPEffi	string	*required* Energy-Efficient Chilled Water Pump (For air conditioning system type of water-cooled chilled water plant, air-cooled chilled water plant, district cooling system) in kW/m³/s input range: 160, 350
	Energy Efficient Condensing Water Pump	CWPEffi	string	*required* Energy Efficient Condensing Water Pump (For air conditioning system type of water-cooled chilled water plant) in kW/m³/s input range: 140, 300
	Energy-Efficient Cooling Tower	CTEffi	string	*required* Energy-Efficient Cooling Tower (For air conditioning system type of water-cooled chilled water plant) in L/s/kW input range: 1.7, 7
	Air Conditioning System Optimisation	ACOpt	string	*required* Air Conditioning System Optimisation (For air conditioning system type of water-cooled chilled water plant, air-cooled chilled water plant) input range: {"Yes"}
	Energy-Efficient Unitary Air-Conditioning System	UnitaryEffi	string	*required* Energy-Efficient Unitary Air-Conditioning System (For air conditioning system type of unitary air-conditioning system) in kW/RT

				input range: 0.6, 1.1
Air Distribution (A/D) System	Use of Variable Speed Drive (VSD) in AHU/FCU	ADType	string	*required* Use of Variable Speed Drive (VSD) in AHU/FCU (For air distribution system type of AHU-CAV and FCU-CAV) input range: {"AHU-VAV", "FCU-VAV"}
	Percentage Use of Alternative Cooling Technologies	PACT	string	*required* Percentage Use of Alternative Cooling Technologies in % input range: 0, 100
	Energy Efficient Air Distribution Fan	ADFanEffi	string	*required* Energy Efficient Air Distribution Fan (For air conditioning system type of water-cooled chilled water plant, air-cooled chilled water plant, district cooling system) in W/CMH input range: 0.05, 0.18 (ExistingADType ==FCU-CAV and ADType==None) 0.2, 0.5 (ExistingADType ==AHU-CAV and ADType==None) 0.3, 0.6 (ExistingADType ==AHU-VAV or ADType==AHU-VAV) 0.06, 0.22 (ExistingADType ==FCU-VAV or ADType==FCU-VAV)
	Demand-Controlled Ventilation of AC Area	DCV	string	*required* Demand-Controlled Ventilation of AC Area (For air conditioning system type of water-cooled chilled water plant, air-cooled chilled water plant, district cooling system) input range: {"Yes"}
	Use of Energy Recovery Ventilation System	ERVEffi	string	*required* Use of Energy Recovery Ventilation System (For air conditioning system type of water-cooled chilled water plant, air-cooled chilled water plant, district cooling system) in % input range: 30, 95
	Percentage Use of Fans to Offset Cooling Load	PFan	string	*required* Percentage Use of Fans to Offset Cooling Load in % input range: 0, 100
Lighting System	LED Efficacy	LEDEfficacy	string	*required* LED Efficacy (Use of energy efficient LED lights) in lm/W

				input range: 70, 200
	Percentage of Use of Energy Efficient LED Lights	PLED	string	*required* Percentage of Use of Energy Efficient LED Lights in % input range: (0, 100
	Smart Lighting Control Percentage of Application	PLigCrl	string	*required* Percentage of Application (Use of smart lighting control) in % input range: (0, 100
	Daylighting Percentage of Application	PDaylight	string	*required* Percentage of Application (Use of daylighting) in % input range: (0, 100
Mechanical Ventilation (MV) System	Energy-Efficient Mechanical Ventilation Fan	MVFanEffi	string	*required* Energy-Efficient Mechanical Ventilation Fan (Fan Efficiency) in W/CMH input range: 0.05, 0.3
	Demand-Controlled Ventilation of MV Area	PMVCrl	string	*required* Demand-Controlled Ventilation of MV Area in % input range: 0, 100
Other System	Use of Smart Plug Load Control	PLC	string	*required* Use of Smart Plug Load Control input range: {“Yes”}
	Use of Energy Efficient Appliances	EfficientApp	string	*required* Use of Energy Efficient Appliances (For building type of residential building) input range: {“Yes”}
	Use of Energy Efficient Hot Water System	HWCOP	string	*required* Use of Energy Efficient Hot Water System (For building type of hotel, healthcare facility, residential building) input range: 2, 5
	Percentage Use of Solar Hot Water Collectors	PSolarHW	string	*required* Percentage Use of Solar Hot Water Collectors (for building type of hotel, healthcare facility, residential building) in % input range: 0, 100
	Use of Energy Efficient Lift	EfficientLift	string	*required* Use of Energy Efficient Lift input range: {“Yes”}
	Use of Energy Efficient Escalator	EfficientEscalator	string	*required* Use of Energy Efficient Escalator input range: {“Yes”}
	Implementation of Building	BEMS	string	*required* Implementation of Building Energy Management

	Energy Management System			System input range: {“Yes”}
	Rooftop Solar Panel Efficiency	PVEffi	string	*required* Rooftop Solar Panel Efficiency in % input range: (0, 100
	Rooftop Solar Panel Area	PVArea	string	*required* Rooftop Solar Panel Area in m^2 input range: 0, ∞)
	BIPV Solar Panel Efficiency	BIPVEffi	string	*required* BIPV Solar Panel Efficiency in % input range: (0, 100
	BIPV Solar Panel Area	BIPVArea	string	*required* BIPV Solar Panel Area in m^2 input range: 0, ∞)

Response

Variable Name	Type	Description
errcode	int	Error code for the API (0: ok, 1: error)
errmsg	string	Message
report	string	Generated building energy efficiency assessment report in Base64 format

Examples

Example Request

```
{
    “ProjectName”: “Project BBB”,
    “ProjectAddress”: “Project Address”,
    “ProjectPostalCode”: “123456”,
    “ProjectCountry”: “Project Country”,
    “ProjectGFA”: “10000”,
    “ProjectStatus”: “Retrofitting Project”,
    “BuildingData”:
        {
            “BuildingInfo”:
                {
                    “BuildingName”: “Building BBB”,
                    “GFA”: “10000”,
                    “BuildingType”: “Office Building”,
                    “Storey”: 10,
                    “YearTOP”: “2025”,
                    “FBArea”: “0”,
                    “SupermarArea”: “0”,
                    “DCArea”: “100”,
                    “Schedule”: “55”,
                    “MonthlyConsumption”: “20000”,
                    “PACArea”: “80”,
```

```
            "ACType": "Water-cooled Chilled Water Plant",
            "ExistingACEffi": "0.75",
            "ExistingADType": "AHU-VAV",
            "ExistingADEffi": "0.25",
            "ExistingPLED": "20",
            "MVArea": "2000"
        }
    ,
    "Facade":
        {
            "PNVArea": "10",
            "WWR": "0.3",
            "RoofU": "0.6",
            "WallU": "0.6",
            "GlassU": "1.9",
            "GlassSHGC": "0.25",
            "FilmU": "4",
            "FilmSHGC": "0.3",
            "FrameU": "0.7",
            "PaintSR": "0.6",
            "Shading": "0.6"
        }
    ,
    "ACSystem":
        {
            "ChillerEffi": "0.52",
            "CHWPEffi": "230",
            "CWPEffi": "210",
            "CTEffi": "5.2",
            "ACOpt": "Yes"
        }
    ,
    "ADSystem":
        {
            "PACT": "20",
            "ADFanEffi": "0.46",
            "DCV": "Yes",
            "ERVEffi": "70",
            "PFan": "50"
        }
    ,
    "LightingSystem":
        {
            "LEDEfficacy": "160",
            "PLED": "90",
            "PLigCrl": "50",
            "PDaylight": "20"
        }
    ,
    "MVSystem":
        {
            "MVFanEffi": "0.15",
            "PMVCrl": "50"
        }
```

```
            ,
            “OtherSystem”
               {
                   “PLC”: “Yes”,
                   “EfficientLift”: “Yes”,
                   “EfficientEscalator”: “Yes”,
                   “BEMS”: “Yes”,
                   “PVEffi”: “19”,
                   “PVArea”: “100”,
                   “BIPVEffi”: “19”,
                   “BIPVArea”: “100”
               }

        }

}
```

Example Response

```
{
    “result”:
        {
            “errcode”: 0,
            “errmsg”: “null”
        },
    “data”:
        {
            “report”: “JVBERi0xLjMKMyAwIG9iago8PC9UeXBlIC9QYWdlCi9QYXJlbnQgMSAwIFIK...”
        }
}
```

2.2.3 In-Operation Project for Building

- Generate a building energy efficiency assessment report of an in-operation project.
- Return API response

URL: https://sleb.sg/slebapi/beea/project/report/inoperation
Method: POST

Request

Category	Data Point	Variable Name	Type	Description
-	Project Name	ProjectName	string	*required* Project name
-	Project Address	ProjectAddress	string	*required* Project address
-	Project Postal Code	ProjectPostalCode	string	*required* Project postal code
-	Project Country	ProjectCountry	string	*required* Project country
-	Project Gross Floor Area	ProjectGFA	string	*required* Project gross floor area (excluding car park area) in m^2 input range: (0, ∞)
-	Project Status	ProjectStatus	string	*required* Project status input range: {"In Operation"}
-	Building Data	BuildingData	object	*required* Building data
-	Building Information	BuildingInfo	object	*required* Building data under category of building information
-	Facade	Facade	object	*required* Building data under category of facade
-	Air Conditioning System	ACSystem	object	*required* Building data under category of air conditioning (A/C) system
-	Air Distribution System	ADSystem	object	*required* Building data under category of air distribution (A/D) system
-	Lighting System	LightingSystem	object	*required* Building data under category of lighting system
-	Mechanical Ventilation System	MVSystem	object	*required* Building data under category of mechanical ventilation (MV) system
-	Other System	OtherSystem	object	*required* Building data under category of other system
Building Information	Building Name	BuildingName	string	*required* Building Name input range: free text
	Building Gross Floor Area	GFA	string	*required* Building Gross Floor Area (excluding car park area) in m^2 input range: (0, ∞)
	Building Type	BuildingType	string	*required* Building Type input range: {"Office Building", "Retail Building", "Hotel", "Healthcare Facility", "Educational

				Institution", "Industrial Building", "Residential Building", "Mixed Development", "Others"}
	Number of Storeys	Storey	int	*required* Number of Storeys input range: 1, 300
	Year of Temporary Occupation Permit or Certificate of Statutory Completion	YearTOP	string	*required* Year of Temporary Occupation Permit or Certificate of Statutory Completion input range: 1900-9999
	Food and Beverage Area	FBArea	string	*required* Food and Beverage Area in m^2 input range: (0, ∞)
	Supermarket Area	SupermarArea	string	*required* Supermarket Area in m^2 input range: (0, ∞)
	Data Centre/Server Room Area	DCArea	string	*required* Data Centre/Server Room Area in m^2 input range: (0, ∞)
	Total Building Operation	Schedule	string	*required* Total Building Operation in hours per week input range: (0, 168
	Average Monthly Energy Consumption	MonthlyConsumption	string	*required* Average Monthly Energy Consumption in kWh input range: (0, ∞)
	Percentage of Air-Conditioned Area	PACArea	string	*required* Percentage of Air-Conditioned Area in % input range: 0, 100
	Air Conditioning System Type	ACType	string	*required* Air Conditioning System Type input range: {"Water-cooled Chilled Water Plant", "Air-cooled Chilled Water Plant", "Unitary Air-conditioning System", "District Cooling System"}
	Air Conditioning System Efficiency	ExistingACEffi	string	*required* Air Conditioning System Efficiency in kW/RT input range: (0, 3
	Air Distribution System Type	ExistingADType	string	*required* Air Distribution System Type input range: {"AHU-CAV", "AHU-VAV", "FCU-CAV", "FCU-VAV"}

	Air Distribution System Efficiency	ExistingADEffi	string	*required* Air Distribution System Efficiency in kW/RT input range: (0, 1
	Total Mechanically Ventilated Area	MVArea	string	*required* Total Mechanically Ventilated Area in m^2 input range: 0, ∞)
	Percentage of Natural Ventilation Area Converted from AC Area	PNVArea	string	*required* Percentage of Natural Ventilation Area Converted from AC Area (Design of natural ventilation) in % input range: 0, PACArea
Facade	Reduced Window to Wall Ratio	WWR	string	*required* Reduced Window to Wall Ratio input range: 0, 0.4
	Roof U-Value (Insulation of roof)	RoofU	string	*required* Roof U-Value (Insulation of roof) in W/m²·K input range: 0.2, 1.5
	Wall U-Value	WallU	string	*required* Wall U-Value (Insulation of external walls) in W/m²·K input range: 0.2, 1.5
	Glass U-Value	GlassU	string	*required* Glass U-Value (Use of high-performance glass) in W/m²·K input range: 0.5, 2.8
	Glass Solar Heat Gain Coefficient	GlassSHGC	string	*required* Solar Heat Gain Coefficient (Use of high-performance glass) input range: 0.1, 0.35
	Solar Window Film U-Value	FilmU	string	*required* Solar Window Film U-Value (Use of solar window film) in W/m²·K input range: 1, 10
	Film Solar Heat Gain Coefficient	FilmSHGC	string	*required* Solar Heat Gain Coefficient (Use of solar window film) input range: 0.05, 0.6
	Window Frame U-Value	FrameU	string	*required* Window Frame U-Value (Use of thermally broken window frames) in W/m²·K input range: 0.4, 1.2
	Solar Reflectivity	PaintSR	string	*required* Solar Reflectivity (Use of reflective paint/cool paint) input range: 0.3, 0.9

	Shading Coefficient	Shading	string	*required* Shading Coefficient (Installation of external shading devices) input range: 0.1, 0.9
	Air Conditioning System Optimisation	ACOpt	string	*required* Air Conditioning System Optimisation (For air conditioning system type of water-cooled chilled water plant, air-cooled chilled water plant) input range: {"Yes"}
Air Conditioning (A/C) System	Energy-Efficient Unitary Air-Conditioning System	UnitaryEffi	string	*required* Energy-Efficient Unitary Air-Conditioning System (For air conditioning system type of unitary air-conditioning system) in kW/RT input range: 0.6, 1.1
	Use of Variable Speed Drive (VSD) in AHU/FCU	ADType	string	*required* Use of Variable Speed Drive (VSD) in AHU/FCU (For air distribution system type of AHU-CAV and FCU-CAV) input range: {"AHU-VAV", "FCU-VAV"}
Air Distribution (A/D) System	Demand-Controlled Ventilation of AC Area	DCV	string	*required* Demand-Controlled Ventilation of AC Area (For air conditioning system type of water-cooled chilled water plant, air-cooled chilled water plant, district cooling system) input range: {"Yes"}
	Use of Energy Recovery Ventilation System	ERVEffi	string	*required* Use of Energy Recovery Ventilation System (For air conditioning system type of water-cooled chilled water plant, air-cooled chilled water plant, district cooling system) in % input range: 30, 95
	Percentage Use of Fans to Offset Cooling Load	PFan	string	*required* Percentage Use of Fans to Offset Cooling Load in % input range: 0, 100
	LED Efficacy	LEDEfficacy	string	*required* LED Efficacy (Use of energy efficient LED lights) in lm/W input range: 70, 200
Lighting System	Percentage of Use of Energy	PLED	string	*required* Percentage of Use of Energy Efficient LED Lights in %

	Efficient LED Lights			input range: (0, 100
	Smart Lighting Control Percentage of Application	PLigCrl	string	*required* Percentage of Application (Use of smart lighting control) in % input range: (0, 100
	Daylighting Percentage of Application	PDaylight	string	*required* Percentage of Application (Use of daylighting) in % input range: (0, 100
	Energy-Efficient Mechanical Ventilation Fan	MVFanEffi	string	*required* Energy-Efficient Mechanical Ventilation Fan (Fan Efficiency) in W/CMH input range: 0.05, 0.3
Mechanical Ventilation (MV) System	Demand-Controlled Ventilation of MV Area	PMVCrl	string	*required* Demand-Controlled Ventilation of MV Area in % input range: 0, 100
	Use of Smart Plug Load Control	PLC	string	*required* Use of Smart Plug Load Control input range: {“Yes”}
Other System	Use of Energy Efficient Appliances	EfficientApp	string	*required* Use of Energy Efficient Appliances (For building type of residential building) input range: {“Yes”}
	Use of Energy Efficient Hot Water System	HWCOP	string	*required* Use of Energy Efficient Hot Water System (For building type of hotel, healthcare facility, residential building) input range: 2, 5
	Percentage Use of Solar Hot Water Collectors	PSolarHW	string	*required* Percentage Use of Solar Hot Water Collectors (for building type of hotel, healthcare facility, residential building) in % input range: 0, 100
	Use of Energy Efficient Lift	EfficientLift	string	*required* Use of Energy Efficient Lift input range: {“Yes”}
	Use of Energy Efficient Escalator	EfficientEscalator	string	*required* Use of Energy Efficient Escalator input range: {“Yes”}
	Implementation of Building Energy Management System	BEMS	string	*required* Implementation of Building Energy Management System input range: {“Yes”}

	Rooftop Solar Panel Efficiency	PVEffi	string	*required* Rooftop Solar Panel Efficiency in % input range: (0, 100
	Rooftop Solar Panel Area	PVArea	string	*required* Rooftop Solar Panel Area in m^2 input range: 0, ∞)
	BIPV Solar Panel Efficiency	BIPVEffi	string	*required* BIPV Solar Panel Efficiency in % input range: (0, 100
	BIPV Solar Panel Area	BIPVArea	string	*required* BIPV Solar Panel Area in m^2 input range: 0, ∞)

Response

Variable Name	Type	Description
errcode	int	Error code for the API (0: ok, 1: error)
errmsg	string	Message
report	string	Generated building energy efficiency assessment report in Base64 format

Examples

Example Request

```
{
    “ProjectName”: “Project CCC”,
    “ProjectAddress”: “Project Address”,
    “ProjectPostalCode”: “123456”,
    “ProjectCountry”: “Project Country”,
    “ProjectGFA”: “10000”,
    “ProjectStatus”: “Retrofitting Project”,
    “BuildingData”:
        {
            “BuildingInfo”:
                {
                    “BuildingName”: “Building CCC”,
                    “GFA”: “10000”,
                    “BuildingType”: “Office Building”,
                    “Storey”: 10,
                    “YearTOP”: “2025”,
                    “FBArea”: “0”,
                    “SupermarArea”: “0”,
                    “DCArea”: “100”,
                    “Schedule”: “55”,
                    “MonthlyConsumption”: “20000”,
                    “PACArea”: “80”,
                    “ACType”: “Water-cooled Chilled Water Plant”,
                    “ExistingACEffi”: “0.75”,
```

```
            "ExistingADType": "AHU-VAV",
            "ExistingADEffi": "0.25",
            "MVArea": "2000",
        }
    ,
    "Facade":
        {
            "PNVArea": "10",
            "WWR": "0.3",
            "RoofU": "0.6",
            "WallU": "0.6",
            "GlassU": "1.9",
            "GlassSHGC": "0.25",
            "FilmU": "4",
            "FilmSHGC": "0.3",
            "FrameU": "0.7",
            "PaintSR": "0.6",
            "Shading": "0.6"
        }
    ,
    "ACSystem":
        {
            "ACOpt": "Yes"
        }
    ,
    "ADSystem":
        {
            "DCV": "Yes",
            "ERVEffi": "70",
            "PFan": "50"
        }
    ,
    "LightingSystem":
        {
            "LEDEfficacy": "160",
            "PLED": "90",
            "PLigCrl": "50",
            "PDaylight": "20"
        }
    ,
    "MVSystem":
        {
            "MVFanEffi": "0.15",
            "PMVCrl": "50"
        }
    ,
    "OtherSystem"
        {
            "PLC": "Yes",
            "EfficientLift": "Yes",
            "EfficientEscalator": "Yes",
            "BEMS": "Yes",
            "PVEffi": "19",
            "PVArea": "100",
```

```
                    “BIPVEffi”: “19”,
                    “BIPVArea”: “100”
                }

        }

}
```


Example Response


```
{
    “result”:
        {
            “errcode”: 0,
            “errmsg”: “null”
        },
    “data”:
        {
            “report”: “JVBERi0xLjMKMyAwIG9iago8PC9UeXBlIC9QYWdlCi9QYXJlbnQgMSAwIFIK...”
        }
}
```

2.2.4 New Development Project for Industrial Facility

- Generate an energy efficiency assessment result of a new development project for an industrial facility.
- Return API response

URL: https://sleb.sg/slebapi/beea/project/report/newdevind
Method: POST

Category	Data Point	Variable Name	Type	Description
-	Project Name	ProjectName	string	*required* Project name
-	Project Address	ProjectAddress	string	*required* Project address
-	Project Postal Code	ProjectPostalCode	string	*required* Project postal code
-	Project Country	ProjectCountry	string	*required* Project country input range: {“Egypt”}
-	Project Gross Floor Area	ProjectGFA	string	*required* Project gross floor area (excluding car park area) in m^2 input range: (0, ∞)
-	Project Status	ProjectStatus	string	*required* Project status input range: {“New Development”}
-	Building Data	BuildingData	object	*required* Building data
-	Building Information	BuildingInfo	object	*required* Building data under category of building information
-	System Types	SystemTypes	array	*required* one or more System Types input range: {"Motors", "Compressed Air Systems", "Boilers", "Furnaces", "Ovens", "Air Conditioners (AC Units)", "Air Handling Fans", "Lighting Systems", "Water Heaters", "Chillers", "Cooling Towers", "Computers / Data Centers", "Elevators / Mechanical Equipment", "Clinker Handling / Conveyors", "Pumps", "Renewable Generation (Solar PV/Thermal/Wind)", "Water/Wastewater Treatment", "Construction Site Energy"}
Building Information	Building Name	BuildingName	string	*required* Building Name input range: free text
	Building Gross Floor Area	GFA	string	*required* Building Gross Floor Area (excluding car

				park area) in m^2 input range: (0, ∞)
	Building Type	BuildingType	string	*required* Building Type input range: {“Industrial Facility”}
	Industry Type	IndustryType	string	*required* Industry Type input range: {“Cement”, “Iron & Steel”, “Fertilizers”, “Chemicals”, “Textiles”, “Food”, “Pharmaceuticals”, “Engineering”, “Automotive”, “Furniture”, “Printing & Packaging”, “Leather”, “Handicraft”, “Hydrocarbons”, “Electricity”, “Construction”, “Electronics”}
	Chemical Product Type	ChemicalProductType	string	*Required* if IndustryType is “Chemicals”; not applicable for other industries. input range: {“Ethylene”, “Methanol”}
	Average Monthly Production Quantity	MonthlyProduction	string	*required* Average Monthly Production Quantity in *ton (Cement, Iron & Steel, Chemicals, Fertilizers, Textiles, Food & Beverage, Printing & Packaging, Leather Industry); piece (Engineering, Automotive, Furniture, Electronics); US$1000 revenue (Pharmaceuticals, Handicraft); boe (Hydrocarbons); kWh (Electricity); m² (Construction).* input range: (0, ∞)
	Primary Fuel Type	PrimaryFuelType	string	*optional* The facility’s expected primary thermal fuel type at this design stage, if known – used only to select a more accurate emission factor for *TotalCarbonEmission*; does not affect *TotalEnergyConsumption*. Omit if not yet decided – the engine falls back to an industry-typical or generic blended estimate. input range: {“Natural

				Gas", "Coal", "Diesel", "Petrol", "LPG", "LNG", "HFO", "LFO", "Electricity"}

Response

Variable Name	Type	Description
errcode	int	Error code for the API (0: ok, 1: error)
errmsg	string	Message
TotalEnergyConsumption	string	Total energy consumption in GJ/year. This is the estimated value based on baseline scenario (i.e., current industrial performance).
TotalCarbonEmission	string	Total carbon emission in t CO2e/year. This is the estimated value based on baseline scenario.
SpecificEnergyConsumptionBenchmarking	object	The Specific Energy Consumption (SEC) - based on baseline scenario, Best Available Techniques (BAT) benchmark, and benchmarking result (expressed as "On Track", i.e., equal and better than BAT, or "Needs Improvement", i.e., worse than BAT). SEC and SECOfBAT share the same unit as the industry's primary energy KPI (see SECUnit). For example, "SEC": "4", "SECOfBAT": "3.4", "SECUnit": "GJ/t clinker", "SECBenchmarkingResult": "Needs Improvement"
KPIBenchmarking	object	The key point indicators specific to the IndustryType selected in the request based on baseline scenario, each with its BAT benchmark and benchmarking result. Each metric uses the same Variable Name as its corresponding "Facility KPIs - <Industry Type>" input parameter in Section 2.2.5 (see there for the unit of each metric), paired with "<Name>OfBAT", "<Name>BenchmarkingResult" and "<Name>Recommendation" keys; "<Name>Recommendation" is always present, shown as "-" when that KPI's BenchmarkingResult is "On Track". The specific KPI fields returned depend on IndustryType. For example, for IndustryType "Iron & Steel":

		“ScrapRecoveryRate”: “90”, “ScrapRecoveryRateOfBAT”: “95”, “ScrapRecoveryRateBenchmarkingResult”: “Needs Improvement”, “ScrapRecoveryRateRecommendation”: [“Improve scrap sorting/grading and magnetic separation to reduce yield losses”]

SystemPerformanceBenchmarking	object	The efficiencies of the input systems - based on baseline scenario, BAT benchmark and benchmarking result, grouped by system type. Each system is keyed by its Variable Name (matching the system objects in Section 2.2.5, e.g. “Furnaces”), and each metric inside uses the same Variable Name as the corresponding system input field (see there for the unit), paired with “<Name>OfBAT” and a shared “BenchmarkingResult” key. Only the systems selected in SystemTypes in the request are included. For example, for SystemTypes including “Furnaces”:
		“Furnaces”: {
		“ThermalEfficiency”: “82”,
		“ThermalEfficiencyOfBAT”: “85”,
		“BenchmarkingResult”: “Needs Improvement”
		}

EnergyAnalysisAndRecommendation	object	Estimated system energy consumption based on baseline scenario, recommended green measures and estimated energy savings, grouped by system type, plus an overall total. Each system is keyed by its Variable Name (e.g. “Furnaces”), containing “EnergyConsumption” and “EnergySavings” (both in GJ) and a “Recommendation” list; “TotalEnergySavings” (GJ) sits alongside the per-system objects. Only the systems selected in SystemTypes in the request are included. For example, for SystemTypes including “Furnaces”: “Furnaces”: { “EnergyConsumption”: “1116290”,

		"Recommendation": ["Optimise combustion air control to reduce excess air ratio", "Install flue gas waste heat recovery system"], "EnergySavings": "167000" }, "TotalEnergySavings": "171940" *Note: For "Renewable Generation (Solar PV/Thermal/Wind)", "EnergyConsumption" is reported as a negative value (GJ) representing the estimated annual generation (offset). "EnergySavings" remains non-negative. Both Renewable Generation and Construction Site Energy (a one-time total, not an annual value) are excluded from "TotalEnergySavings".*
WaterAnalysisAndRecommendation	object	Water consumption (m^3) based on baseline scenario, recommended green measures ("WaterRecommendation") and estimated water savings ("TotalWaterSavings", m^3), where applicable. For example, "WaterConsumption": "100000", "WaterRecommendation": ["Recover condensate"], "TotalWaterSavings": "20000"
RawMaterialAnalysisAndRecommendation	object	Raw material consumption (ton) based on baseline scenario, recommended green measures ("RawMaterialRecommendation") and estimated raw material savings ("TotalRawMaterialSavings", ton), where applicable. For example, "RawMaterialConsumption": "1000", "RawMaterialRecommendation": ["Continuous casting optimization"], "TotalRawMaterialSavings": "50"

Examples

Example Request

```
{

    "ProjectName": "AAA",
    "ProjectAddress": "BBB",
    "ProjectPostalCode": "000000",
    "ProjectCountry": "Egypt",
    "ProjectGFA": "18000",
    "ProjectStatus": "New Development",
    "BuildingData":
        {
            "BuildingInfo":
                {
                    "BuildingName": "CCC",
```

```
                    "GFA": "18000",
                    "BuildingType": "Industrial Facility",
                    "IndustryType": "Automotive",
                    "MonthlyProduction": "4500",
                    "PrimaryFuelType": "Natural Gas"
                }
            ,
            "SystemTypes": ["Motors", "Air Conditioners (AC Units)", "Lighting Systems"]
        }
}
```


Example Response


```
{
    "result":
        {
            "errcode": 0,
            "errmsg": ""
        }
    ,
    "data":
        {
            "TotalEnergyConsumption": "297000",
            "TotalCarbonEmission": "16661.7",
            "SpecificEnergyConsumptionBenchmarking":
                {
                    "SEC": "5.5",
                    "SECOfBAT": "3.5",
                    "SECUnit": "GJ/vehicle",
                    "SECBenchmarkingResult": "Needs Improvement"
                }
            ,
            "KPIBenchmarking":
                {
                    "SpecificEnergyConsumption": "5.5",
                    "SpecificEnergyConsumptionOfBAT": "3.5",
                    "SpecificEnergyConsumptionBenchmarkingResult": "Needs Improvement",
                    “SpecificEnergyConsumptionRecommendation”: [“Install VFDs on all assembly-line conveyors”, “Programme robotic standby/sleep mode during line stoppages”, “Add energy sub-metering per production line”],
                    "PaintShopEnergyIntensity": "290",
                    "PaintShopEnergyIntensityOfBAT": "180",
                    "PaintShopEnergyIntensityBenchmarkingResult": "Needs Improvement",
                    “PaintShopEnergyIntensityRecommendation”: [“Upgrade to a 3-wet paint process with waterborne coatings”, “Install waste-heat recovery (WHR) from the paint-bake oven exhaust”, “Optimise flash-off time between coating stages”],
                    "WaterConsumption": "6.5",
                    "WaterConsumptionOfBAT": "3",
                    "WaterConsumptionBenchmarkingResult": "Needs Improvement",
                    “WaterConsumptionRecommendation”: [“Install closed-loop water recirculation in paint/wash processes”, “Recover heat from wash-tank discharge”, “Target zero liquid discharge (ZLD)”],
                    "CO2Intensity": "0.75",
                    "CO2IntensityOfBAT": "0.4",
```

```
            "CO2IntensityBenchmarkingResult": "Needs Improvement",
            “CO2IntensityRecommendation”: [“Install rooftop/carport solar PV (5-10 MW per assembly plant), optionally with battery storage for peak shaving”, “Combine with paint-shop and assembly-line electrification/efficiency measures above to cut grid-linked CO2”],
            "ScrapRecoveryRate": "91",
            "ScrapRecoveryRateOfBAT": "97",
            "ScrapRecoveryRateBenchmarkingResult": "Needs Improvement",
            “ScrapRecoveryRateRecommendation”: [“Optimise stamping-die design to reduce metal scrap”],
            "RobotUtilisationRate": "75",
            "RobotUtilisationRateOfBAT": "90",
            "RobotUtilisationRateBenchmarkingResult": "Needs Improvement",
            “RobotUtilisationRateRecommendation”: [“Optimise production scheduling and changeover procedures to raise robot/line utilisation”, “Implement predictive maintenance to reduce unplanned robot downtime”],
            "TyreRollingResistance": "8.2",
            "TyreRollingResistanceOfBAT": "9",
            "TyreRollingResistanceBenchmarkingResult": "On Track",
            “TyreRollingResistanceRecommendation”: “-”
        }
    ,
    "SystemPerformanceBenchmarking":
        {
            "Motors":
                {
                    "MotorEfficiency": "70",
                    "MotorEfficiencyOfBAT": "90",
                    "BenchmarkingResult": "Needs Improvement"
                }
            ,
            "AirConditioners":
                {
                    "EfficiencyValue": "2.7",
                    "EfficiencyValueOfBAT": "4.5",
                    "BenchmarkingResult": "Needs Improvement"
                }
            ,
            "LightingSystems":
                {
                    "LightingPowerDensity": "18",
                    "LightingPowerDensityOfBAT": "8",
                    "BenchmarkingResult": "Needs Improvement"
                }
        }
    ,
    "EnergyAnalysisAndRecommendation":
        {
            "Motors":
                {
                    "EnergyConsumption": "135685.26",
                    "Recommendation": ["Install VFDs on all fans >15 kW", "Upgrade to motors with IE3/IE4", "Audit motor loading and right-sizing; verify VFD control parameters are correctly tuned to the actual duty cycle"],
```

```
                        "EnergySavings": "42876.54"
                    }
                ,
                "AirConditioners":
                    {
                        "EnergyConsumption": "120609.16",
                        "Recommendation": ["Implement a filter/coil maintenance programme",
                        "Replace fixed-speed split/packaged units with inverter-driven, high-EER
                        (>4.5) systems", "Install programmable/BMS-based scheduling and
                        setback controls to avoid conditioning unoccupied periods", "Conduct a
                        broader AC system audit (refrigerant charge, coil condition,
                        ductwork/airflow) where maintenance and controls are already in good
                        standing but a performance gap remains"],
                        "EnergySavings": "47143.71"
                    }
                ,
                "LightingSystems":
                    {
                        "EnergyConsumption": "40705.58",
                        "Recommendation": ["Install occupancy and daylight sensors with
                        centralised lighting controls", "Replace mercury-vapour/HPS/fluorescent
                        lamps with LED fixtures", "Audit lighting fixture output and maintenance
                        factor; verify occupancy/daylight controls are functioning as designed
                        and re-commission sensors and scheduling"],
                        "EnergySavings": "25914.19"
                    }
                ,
                "TotalEnergySavings": "115934.44"
            }
        ,
        "WaterAnalysisAndRecommendation":
            {
                "WaterConsumption": "351000",
                "WaterRecommendation": ["Install closed-loop water recirculation in
                paint/wash processes", "Recover heat from wash-tank discharge", "Target zero
                liquid discharge (ZLD)"],
                "TotalWaterSavings": "202921.88"
            }
        ,
        "RawMaterialAnalysisAndRecommendation": null
    }
}
```

2.2.5 Existing Project for Industrial Facility

- Generate an energy efficiency assessment result of an existing industrial facility.
- Return API response

URL: https://sleb.sg/slebapi/beea/project/report/existingind
Method: POST

Category	Data Point	Variable Name	Type	Description
-	Project Name	ProjectName	string	*required* Project name
-	Project Address	ProjectAddress	string	*required* Project address
-	Project Postal Code	ProjectPostalCode	string	*required* Project postal code
-	Project Country	ProjectCountry	string	*required* Project country input range: {“Egypt”}
-	Project Gross Floor Area	ProjectGFA	string	*required* Project gross floor area (excluding car park area) in m^2 input range: (0, ∞)
-	Project Status	ProjectStatus	string	*required* Project status input range: {“Existing Building”}
-	Building Data	BuildingData	object	*required* Building data
-	Building Information	BuildingInfo	object	*required* Building data under category of building information
-	System Types	SystemTypes	array	*required* one or more System Types input range: {"Motors", "Compressed Air Systems", "Boilers", "Furnaces", "Ovens", "Air Conditioners (AC Units)", "Air Handling Fans", "Lighting Systems", "Water Heaters", "Chillers", "Cooling Towers", "Computers / Data Centers", "Elevators / Mechanical Equipment", "Clinker Handling / Conveyors", "Pumps", "Renewable Generation (Solar PV/Thermal/Wind)", "Water/Wastewater Treatment", "Construction Site Energy"} *Note: where a building has multiple pieces of equipment of the same system type (e.g., several motors), provide the aggregate/overall value for that system in the respective system object below- for example, combined rated power and a production-*

				weighted average efficiency - rather than a value for a single unit.
-	Motors	Motors	object	*Required* if "Motors" is selected in SystemTypes; contains motor assessment data.
-	Compressed Air Systems	CompressedAirSystems	object	*Required* if "Compressed Air Systems" is selected in SystemTypes; contains compressed air system assessment data.
-	Boilers	Boilers	object	*Required* if "Boilers" is selected in SystemTypes; contains boiler assessment data.
-	Furnaces	Furnaces	object	*Required* if "Furnaces" is selected in SystemTypes; contains furnace assessment data.
-	Ovens	Ovens	object	*Required* if "Ovens" is selected in SystemTypes; contains oven assessment data.
-	Air Conditioners (AC Units)	AirConditioners	object	*Required* if "Air Conditioners (AC Units)" is selected in SystemTypes; contains air conditioner assessment data.
-	Air Handling Fans	AirHandlingFans	object	*Required* if "Air Handling Fans" is selected in SystemTypes; contains air handling fan assessment data.
-	Lighting Systems	LightingSystems	object	*Required* if "Lighting Systems" is selected in SystemTypes; contains lighting system assessment data.
-	Water Heaters	WaterHeaters	object	*Required* if "Water Heaters" is selected in SystemTypes; contains water heater assessment data.
-	Chillers	Chillers	object	*Required* if "Chillers" is selected in SystemTypes; contains chiller assessment data.
-	Cooling Towers	CoolingTowers	object	*Required* if "Cooling Towers" is selected in SystemTypes; contains cooling tower assessment data.
-	Computers / Data Centers	ComputersDataCenters	object	*Required* if "Computers / Data Centers" is selected in SystemTypes; contains computer and data centre assessment data.
-	Elevators / Mechanical Equipment	ElevatorsMechanicalEquipment	object	*Required* if "Elevators / Mechanical Equipment" is selected in SystemTypes; contains elevator and mechanical equipment assessment data.
-	Clinker Handling / Conveyors	ClinkerHandlingConveyors	object	*Required* if "Clinker Handling / Conveyors" is selected in

				SystemTypes; contains clinker handling and conveyor assessment data.
-	Pumps	Pumps	object	*Required* if “Pumps” is selected in SystemTypes; contains pump assessment data.
-	Renewable Generation (Solar PV/Thermal/Wind)	RenewableGeneration	object	*Required* if “Renewable Generation (Solar PV/Thermal/Wind)” is selected in SystemTypes; contains renewable generation assessment data.
-	Water/Wastewater Treatment	WaterWastewaterTreatment	object	*Required* if “Water/Wastewater Treatment” is selected in SystemTypes; contains water/wastewater treatment assessment data.
-	Construction Site Energy	ConstructionSiteEnergy	object	*Required* if "Construction Site Energy" is selected in SystemTypes; contains construction site energy assessment data.
Building Information	Building Name	BuildingName	string	*required* Building Name input range: free text
	Building Gross Floor Area	GFA	string	*required* Building Gross Floor Area (excluding car park area) in m^2 input range: (0, ∞)
	Building Type	BuildingType	string	*required* Building Type input range: {“Industrial Facility”}
	Industry Type	IndustryType	string	*required* Industry Type input range: {“Cement”, “Iron & Steel”, “Fertilizers”, “Chemicals”, “Textiles”, “Food”, “Pharmaceuticals”, “Engineering”, “Automotive”, “Furniture”, “Printing & Packaging”, “Leather”, “Handicraft”, “Hydrocarbons”, “Electricity”, “Construction”, “Electronics”} *Note: for “Electricity”, represent the generation/steam-cycle train using “Boilers” (e.g., HRSG or steam generation).*
	Chemical Product Type	ChemicalProductType	string	*Required* if IndustryType is “Chemicals”; not applicable for other industries. input range: {“Ethylene”, “Methanol”}

	Average Monthly Production Quantity	MonthlyProduction	string	*required* Average Monthly Production Quantity in *ton (Cement, Iron & Steel, Chemicals, Fertilizers, Textiles, Food & Beverage, Printing & Packaging, Leather Industry); piece (Engineering, Automotive, Furniture, Electronics); US$1000 revenue (Pharmaceuticals, Handicraft); boe (Hydrocarbons); kWh (Electricity); m² (Construction).* input range: (0, ∞)
	Consumption	Consumption	object	*required* Consumption data under category of Consumption
	Facility KPIs	FacilityKPIs	object	*required* Facility KPI data under category of Facility KPIs - <Industry Type>
Consumption	Average Monthly Electricity Consumption	MonthlyElectricityConsumption	string	*required* Average Monthly Electricity Consumption in kWh input range: [0, ∞)
	Average Monthly Natural Gas Consumption	MonthlyNaturalGasConsumption	string	*required* Average Monthly Natural Gas Consumption in m3 input range: [0, ∞)
	Average Monthly Coal Consumption	MonthlyCoalConsumption	string	*required* Average Monthly Coal Consumption in ton input range: [0, ∞)
	Average Monthly Diesel Consumption	MonthlyDieselConsumption	string	*required* Average Monthly Diesel Consumption in m3 input range: [0, ∞)
	Average Monthly Petrol Consumption	MonthlyPetrolConsumption	string	*required* Average Monthly Petrol Consumption in m3 input range: [0, ∞)
	Average Monthly LPG Consumption	MonthlyLPGConsumption	string	*required* Average Monthly LPG Consumption in ton input range: [0, ∞)
	Average Monthly LNG Consumption	MonthlyLNGConsumption	string	*required* Average Monthly LNG Consumption in ton input range: [0, ∞)
	Average Monthly HFO Consumption	MonthlyHFOConsumption	string	*required* Average Monthly HFO Consumption in m3 input range: [0, ∞)
	Average Monthly LFO Consumption	MonthlyLFOConsumption	string	*required* Average Monthly LFO Consumption in m3 input range: [0, ∞)

	Average Monthly Other Energy Consumption	MonthlyOtherConsumption	string	*required* Average Monthly Other Consumption in GJ input range: [0, ∞)
	Weighted Emission Factor of Other Energy	OtherEnergyEF	string	*required* Weighted Emission Factor of Other Energy in ton CO2e/GJ input range: [0, ∞)
	Average Monthly Water Consumption	MonthlyWaterConsumption	string	*required* Average Monthly Water Consumption in m3 input range: [0, ∞)
	Average Monthly Raw Material Consumption	MonthlyRawMaterialConsumption	string	*required* Average Monthly Raw Material Consumption in ton input range: [0, ∞)
Facility KPIs - Cement	Specific Thermal Energy Consumption	SpecificThermalEnergyConsumption	string	*required* Specific Thermal Energy Consumption in GJ/t clinker input range: (0, ∞)
	Specific Electrical Energy Consumption	SpecificElectricalEnergyConsumption	string	*required* Specific Electrical Energy Consumption in kWh/t cement input range: (0, ∞)
	CO_2 Intensity	CO2Intensity	string	*required* CO_2 Intensity in kg CO_2/t cement input range: (0, ∞)
	Clinker Ratio	ClinkerRatio	string	*required* Clinker Ratio input range: (0, 1]
	Alternative Fuel Rate	AlternativeFuelRate	string	*required* Alternative Fuel Rate in % input range: [0, 100]
Facility KPIs - Iron & Steel	Specific Energy Consumption (EAF route)	SpecificEnergyConsumptionEAFRoute	string	*required* Specific Energy Consumption (EAF route) in GJ/t crude steel input range: (0, ∞)
	EAF Electricity Intensity	EAFElectricityIntensity	string	*required* EAF Electricity Intensity in kWh/t crude steel input range: (0, ∞)
	Specific Energy Consumption (DRI–EAF route)	SpecificEnergyConsumptionDRIEAFRoute	string	*required* Specific Energy Consumption (DRI–EAF route) in GJ/t crude steel input range: (0, ∞)
	Motor System Efficiency	MotorSystemEfficiency	string	*required* Motor System Efficiency in % input range: [0, 100]

	CO_2 Intensity	CO2Intensity	string	*required* CO_2 Intensity in t CO_2/t steel input range: (0, ∞)
	Scrap Recovery Rate	ScrapRecoveryRate	string	*required* Scrap Recovery Rate in % input range: [0, 100]
Facility KPIs - Chemicals	Specific Energy Consumption	SpecificEnergyConsumption	string	*required* Specific Energy Consumption in GJ/ton product input range: (0, ∞)
	Motor System Efficiency	MotorSystemEfficiency	string	*required* Motor System Efficiency in % input range: [0, 100]
	Compressed Air Specific Energy	CompressedAirSpecificEnergy	string	*required* Compressed Air Specific Energy in kWh/m³ compressed air input range: (0, ∞)
	Steam System Efficiency	SteamSystemEfficiency	string	*required* Steam System Efficiency in % input range: [0, 100]
	Process Heat Recovery Rate	ProcessHeatRecoveryRate	string	*required* Process Heat Recovery Rate in % input range: [0, 100]
Facility KPIs - Fertilizers	Ammonia Energy Intensity	AmmoniaEnergyIntensity	string	*required* Ammonia Energy Intensity in GJ/t NH_3 input range: (0, ∞)
	Natural Gas Consumption Intensity	NaturalGasConsumptionIntensity	string	*required* Natural Gas Consumption Intensity in Nm^3/t NH_3 input range: (0, ∞)
	Steam Consumption	SteamConsumption	string	*required* Steam Consumption in GJ/t product input range: (0, ∞)
	CO_2 Capture Rate	CO2CaptureRate	string	*required* CO_2 Capture Rate in % input range: [0, 100]
	Electricity Intensity	ElectricityIntensity	string	*required* Electricity Intensity in kWh/t NH_3 input range: (0, ∞)
Facility KPIs - Textiles	Steam Consumption	SteamConsumption	string	*required* Steam Consumption in GJ/t fabric input range: (0, ∞)
	Electricity Intensity	ElectricityIntensity	string	*required* Electricity Intensity in kWh/t fabric input range: (0, ∞)

	Boiler Efficiency	BoilerEfficiency	string	*required* Boiler Efficiency in % input range: [0, 100]
	Water Consumption	WaterConsumption	string	*required* Water Consumption in m^3/t fabric input range: (0, ∞)
	Dyebath Heat Recovery Rate	DyebathHeatRecoveryRate	string	*required* Dyebath Heat Recovery Rate in % input range: [0, 100]
Facility KPIs - Food	Thermal Energy Intensity	ThermalEnergyIntensity	string	*required* Thermal Energy Intensity in GJ/t product input range: (0, ∞)
	Electricity Intensity	ElectricityIntensity	string	*required* Electricity Intensity in kWh/t product input range: (0, ∞)
	Refrigeration COP	RefrigerationCOP	string	*required* Refrigeration COP in COP input range: (0, ∞)
	Boiler Efficiency	BoilerEfficiency	string	*required* Boiler Efficiency in % input range: [0, 100]
	Renewable Energy Share	RenewableEnergyShare	string	*required* Renewable Energy Share in % input range: [0, 100]
Facility KPIs - Pharmaceuticals	Energy Intensity	EnergyIntensity	string	*required* Energy Intensity in GJ / $M revenue input range: (0, ∞)
	Electricity Consumption	ElectricityConsumption	string	*required* Electricity Consumption in kWh/m^2/yr input range: (0, ∞)
	Water Consumption Intensity	WaterConsumptionIntensity	string	*required* Water Consumption Intensity in m^3/t product input range: (0, ∞)
	CO_2 Intensity	CO2Intensity	string	*required* CO_2 Intensity in kg CO_2/$M revenue input range: (0, ∞)
	Waste Generation Rate	WasteGenerationRate	string	*required* Waste Generation Rate in kg waste/t product input range: (0, ∞)
	Solvent Recovery Rate	SolventRecoveryRate	string	*required* Solvent Recovery Rate in % input range: [0, 100]
	Specific Energy Consumption	SpecificEnergyConsumption	string	*required* Specific Energy Consumption in kWh/unit

				input range: (0, ∞)
	Motor System Efficiency	MotorSystemEfficiency	string	*required* Motor System Efficiency in % input range: [0, 100]
	Compressed Air Efficiency	CompressedAirEfficiency	string	*required* Compressed Air Efficiency in kWh/m³ input range: (0, ∞)
	Material Utilisation Rate	MaterialUtilisationRate	string	*required* Material Utilisation Rate in % input range: [0, 100]
Facility KPIs - Engineering	Defect Rate (PPM)	DefectRatePPM	string	*required* Defect Rate (PPM) in PPM input range: (0, ∞)
	CO_2 Intensity	CO2Intensity	string	*required* CO_2 Intensity in kg CO_2/t product input range: (0, ∞)
Facility KPIs - Automotive	Specific Energy Consumption	SpecificEnergyConsumption	string	*required* Specific Energy Consumption in GJ/vehicle input range: (0, ∞)
	Paint Shop Energy Intensity	PaintShopEnergyIntensity	string	*required* Paint Shop Energy Intensity in kWh/vehicle input range: (0, ∞)
	Water Consumption	WaterConsumption	string	*required* Water Consumption in m³/vehicle input range: (0, ∞)
	CO_2 Intensity	CO2Intensity	string	*required* CO_2 Intensity in t CO_2/vehicle input range: (0, ∞)
	Scrap Recovery Rate	ScrapRecoveryRate	string	*required* Scrap Recovery Rate in % input range: [0, 100]
	Robot Utilisation Rate	RobotUtilisationRate	string	*required* Robot Utilisation Rate in % input range: [0, 100]
	Tyre Rolling Resistance	TyreRollingResistance	string	*required* Tyre Rolling Resistance in N/kN input range: (0, ∞)
Facility KPIs - Furniture	Wood Material Yield	WoodMaterialYield	string	*required* Wood Material Yield in % input range: [0, 100]
	Specific Energy Consumption	SpecificEnergyConsumption	string	*required* Specific Energy Consumption in kWh/m³

				input range: (0, ∞)
	VOC Emissions Rate	VOCEmissionsRate	string	*required* VOC Emissions Rate in g VOC/m² surface input range: (0, ∞)
	Dust Extraction Efficiency	DustExtractionEfficiency	string	*required* Dust Extraction Efficiency in % input range: [0, 100]
	Water-based Coatings Share	WaterBasedCoatingsShare	string	*required* Water-based Coatings Share in % input range: [0, 100]
	Waste Wood Recovery Rate	WasteWoodRecoveryRate	string	*required* Waste Wood Recovery Rate in % input range: [0, 100]
Facility KPIs - Printing & Packaging	Specific Energy Consumption	SpecificEnergyConsumption	string	*required* Specific Energy Consumption in kWh/t output input range: (0, ∞)
	Ink & Chemical Waste Rate	InkChemicalWasteRate	string	*required* Ink & Chemical Waste Rate in kg/t product input range: (0, ∞)
	Paper/Material Waste Rate	PaperMaterialWasteRate	string	*required* Paper/Material Waste Rate in % input range: [0, 100]
	VOC Emissions	VOCEmissions	string	*required* VOC Emissions in g VOC/m² printed input range: (0, ∞)
	Recycled Content Share	RecycledContentShare	string	*required* Recycled Content Share in % input range: [0, 100]
	Water Consumption	WaterConsumption	string	*required* Water Consumption in m³/t output input range: (0, ∞)
Facility KPIs - Leather	Water Consumption Intensity	WaterConsumptionIntensity	string	*required* Water Consumption Intensity in m³/t hide input range: (0, ∞)
	Chromium Recovery Rate	ChromiumRecoveryRate	string	*required* Chromium Recovery Rate in % input range: [0, 100]
	Chemical Oxygen Demand (COD)	ChemicalOxygenDemandCOD	string	*required* Chemical Oxygen Demand (COD) in mg/L in effluent input range: (0, ∞)
	Solid Waste Generation	SolidWasteGeneration	string	*required* Solid Waste Generation in kg/t hide

				input range: (0, ∞)
	Energy Consumption	EnergyConsumption	string	*required* Energy Consumption in GJ/t finished leather input range: (0, ∞)
	Effluent Sulphide Content	EffluentSulphideContent	string	*required* Effluent Sulphide Content in mg/L input range: (0, ∞)
Facility KPIs - Handicraft	Energy Intensity	EnergyIntensity	string	*required* Energy Intensity in kWh/$1K revenue input range: (0, ∞)
	Renewable Energy Share	RenewableEnergyShare	string	*required* Renewable Energy Share in % input range: [0, 100]
	Material Waste Rate	MaterialWasteRate	string	*required* Material Waste Rate in % input range: [0, 100]
	Water Consumption	WaterConsumption	string	*required* Water Consumption in m^3/t product input range: (0, ∞)
	Chemical Use Intensity	ChemicalUseIntensity	string	*required* Chemical Use Intensity in kg/t product input range: (0, ∞)
Facility KPIs - Hydrocarbons	Flaring Intensity	FlaringIntensity	string	*required* Flaring Intensity in m^3 gas flared/boe input range: (0, ∞)
	Energy Intensity (Refining)	EnergyIntensityRefining	string	*required* Energy Intensity (Refining) in GJ/barrel input range: (0, ∞)
	GHG Intensity	GHGIntensity	string	*required* GHG Intensity in kg CO_2e/boe input range: (0, ∞)
	Methane Leak Rate	MethaneLeakRate	string	*required* Methane Leak Rate in % input range: [0, 100]
	Water Re-injection Rate	WaterReInjectionRate	string	*required* Water Re-injection Rate in % input range: [0, 100]
	LNG Boil-off Rate	LNGBoilOffRate	string	*required* LNG Boil-off Rate in % per day input range: (0, ∞)
	Refinery Complexity Index	RefineryComplexityIndex	string	*required* Refinery Complexity Index in Index

				input range: (0, ∞)
Facility KPIs - Electricity	Transmission & Distribution Loss	TransmissionDistributionLoss	string	*required* Transmission & Distribution Loss in % input range: [0, 100]
	Net Heat Rate (Thermal)	NetHeatRateThermal	string	*required* Net Heat Rate (Thermal) in kJ/kWh input range: (0, ∞)
	Capacity Factor (Solar PV)	CapacityFactorSolarPV	string	*required* Capacity Factor (Solar PV) in % input range: [0, 100]
	Capacity Factor (Wind)	CapacityFactorWind	string	*required* Capacity Factor (Wind) in % input range: [0, 100]
	Renewable Energy Share	RenewableEnergyShare	string	*required* Renewable Energy Share in % input range: [0, 100]
	Plant Availability Factor	PlantAvailabilityFactor	string	*required* Plant Availability Factor in % input range: [0, 100]
	CO_2 Intensity (Grid)	CO2IntensityGrid	string	*required* CO_2 Intensity (Grid) in g CO_2/kWh input range: (0, ∞)
	SAIDI	SAIDI	string	*required* SAIDI in hours/customer/yr input range: (0, ∞)
Facility KPIs - Construction	Energy Use Intensity (EUI)	EnergyUseIntensityEUI	string	*required* Energy Use Intensity (EUI) in kWh/m²/yr input range: (0, ∞)
	Embodied Carbon Intensity	EmbodiedCarbonIntensity	string	*required* Embodied Carbon Intensity in kg CO_2e/m^2 input range: (0, ∞)
	Construction Waste Rate	ConstructionWasteRate	string	*required* Construction Waste Rate in kg/m² input range: (0, ∞)
	Water Consumption on Site	WaterConsumptionOnSite	string	*required* Water Consumption on Site in m³/m² input range: (0, ∞)
	Concrete Cement Content	ConcreteCementContent	string	*required* Concrete Cement Content in kg/m³ concrete input range: (0, ∞)
	Recycled Material Share	RecycledMaterialShare	string	*required* Recycled Material Share in %

				input range: [0, 100]
	Building Energy Rating	BuildingEnergyRating	string	*required* Building Energy Rating (Star / LEED Level) input range: {“Certified”, “Silver”, “Gold”, “Platinum”, “1 Star”, “2 Star”, “3 Star”, “4 Star”, “5 Star”}
Facility KPIs - Electronics	Specific Energy Consumption	SpecificEnergyConsumption	string	*required* Specific Energy Consumption in kWh/unit input range: (0, ∞)
	Yield Rate	YieldRate	string	*required* Yield Rate in % input range: [0, 100]
	Soldering Defect Rate	SolderingDefectRate	string	*required* Soldering Defect Rate in DPMO input range: (0, ∞)
	Hazardous Waste Generation	HazardousWasteGeneration	string	*required* Hazardous Waste Generation in kg/t output input range: (0, ∞)
	Water Consumption	WaterConsumption	string	*required* Water Consumption in L/m^2 input range: (0, ∞)
	ESD Incident Rate	ESDIncidentRate	string	*required* ESD Incident Rate in per million units input range: (0, ∞)
	OEE (Overall Equipment Eff.)	OEEOverallEquipmentEff	string	*required* OEE (Overall Equipment Eff.) in % input range: [0, 100]
Motors	Rated Power	RatedPower	string	*optional* Rated Power in kW input range: (0, ∞)
	Motor Efficiency	MotorEfficiency	string	*optional* Motor Efficiency in % input range: [0, 100]
	Operating Hours	OperatingHours	string	*optional* Operating Hours in h/yr input range: (0, 8760]
	Load Factor	LoadFactor	string	*optional* Load Factor in % input range: [0, 100]
	VFD Availability	VFDAvailability	string	*optional* VFD Availability input range: {“Yes”, “No”}
	Energy Consumption (Measured)	EnergyConsumptionMeasured	string	*optional* Energy Consumption (Measured) in kWh/yr input range: (0, ∞)
	Rated Capacity	RatedCapacity	string	*optional* Rated Capacity in m^3/min

Compressed Air Systems				input range: (0, ∞)
	Specific Power Consumption	SpecificPowerConsumption	string	*optional* Specific Power Consumption in kW/(m³/min) input range: (0, ∞)
	Operating Hours	OperatingHours	string	*optional* Operating Hours in h/yr input range: (0, 8760]
	Air Leakage Rate	AirLeakageRate	string	*optional* Air Leakage Rate in % input range: [0, 100]
	VFD Availability	VFDAvailability	string	*optional* VFD Availability input range: {“Yes”, “No”}
	Energy Consumption (Measured)	EnergyConsumptionMeasured	string	*optional* Energy Consumption (Measured) in kWh/yr input range: (0, ∞)
Boilers	Rated Capacity	RatedCapacity	string	*optional* Rated Capacity (see Rated Capacity Unit for the unit used) input range: (0, ∞)
	Rated Capacity Unit	RatedCapacityUnit	string	*optional* Rated Capacity Unit input range: {“MW”, “ton steam/hr”}
	Thermal Efficiency	ThermalEfficiency	string	*optional* Thermal Efficiency in % input range: [0, 100]
	Fuel Type	FuelType	string	*optional* Fuel Type input range: {“Natural Gas”, “Diesel”, “Coal”, “Biomass”, “LPG”, “Other”}
	Fuel Energy Input	FuelEnergyInput	string	*optional* Fuel Energy Input in GJ/yr (total fuel energy delivered to the boiler, from fuel purchase/metering records) input range: (0, ∞)
	Boiler Loading	BoilerLoading	string	*optional* Boiler Loading in % input range: [0, 100]
	Operating Hours	OperatingHours	string	*optional* Operating Hours in h/yr input range: (0, 8760]
	Excess Air Level	ExcessAirLevel	string	*optional* Excess Air Level in % input range: [0, 100]
	Condensate Recovery	CondensateRecovery	string	*optional* Condensate Recovery in % input range: [0, 100]
	Energy Consumption (Measured)	EnergyConsumptionMeasured	string	*optional* Energy Consumption (Measured) in GJ/yr input range: (0, ∞)

Furnaces	Rated Capacity	RatedCapacity	string	*optional* Rated Capacity in MW input range: (0, ∞)
	Load Factor	LoadFactor	string	*optional* Load Factor in % input range: [0, 100]
	Operating Hours	OperatingHours	string	*optional* Operating Hours in h/yr input range: (0, 8760]
	Thermal Efficiency	ThermalEfficiency	string	*optional* Thermal Efficiency in % input range: [0, 100]
	Excess Air Ratio	ExcessAirRatio	string	*optional* Excess Air Ratio in ratio input range: (1, ∞)
	Flue Gas Losses	FlueGasLosses	string	*optional* Flue Gas Losses in % of energy input input range: [0, 100]
	Insulation Surface Temperature	InsulationSurfaceTemperature	string	*optional* Insulation Surface Temperature in °C (measured shell/casing temperature; higher values indicate degraded insulation) input range: (0, ∞)
	Energy Consumption (Measured)	EnergyConsumptionMeasured	string	*optional* Energy Consumption (Measured) in kWh/yr input range: (0, ∞)
Ovens	Energy Consumption (Measured)	EnergyConsumptionMeasured	string	*optional* Energy Consumption (Measured) in kWh/yr input range: (0, ∞)
	Rated Capacity	RatedCapacity	string	*optional* Rated Capacity in MW input range: (0, ∞)
	Load Factor	LoadFactor	string	*optional* Load Factor in % input range: [0, 100]
	Operating Hours	OperatingHours	string	*optional* Operating Hours in h/yr input range: (0, 8760]
	Thermal Efficiency	ThermalEfficiency	string	*optional* Thermal Efficiency in % input range: [0, 100]
	Heat Losses through Walls/Openings	HeatLossesThroughWallsOpenings	string	*optional* Heat Losses through Walls/Openings in % of input energy input range: [0, 100]
Air Conditioners (AC Units)	Cooling Capacity	CoolingCapacity	string	*optional* Cooling Capacity (see Cooling Capacity Unit for the unit used)

				input range: (0, ∞)
	Cooling Capacity Unit	CoolingCapacityUnit	string	*optional* Cooling Capacity Unit input range: {“TR”, “kW”}
	Efficiency Value	EfficiencyValue	string	*optional* Efficiency Value (see Efficiency Metric Type for which metric this is) input range: (0, ∞)
	Efficiency Metric Type	EfficiencyMetricType	string	*optional* Efficiency Metric Type input range: {“COP”, “EER”, “SEER”}
	Operating Hours	OperatingHours	string	*optional* Operating Hours in h/yr input range: (0, 8760]
	Months Since Last Service	MonthsSinceLastService	string	*optional* Months Since Last Service input range: [0, ∞)
	Control System	ControlSystem	string	*optional* Control System input range: {“Manual”, “Programmable Thermostat”, “BMS”, “None”}
	Energy Consumption (Measured)	EnergyConsumptionMeasured	string	*optional* Energy Consumption (Measured) in kWh/yr input range: (0, ∞)
Air Handling Fans	Fan Power	FanPower	string	*optional* Fan Power in kW input range: (0, ∞)
	Fan Efficiency	FanEfficiency	string	*optional* Fan Efficiency in % input range: [0, 100]
	Operating Hours	OperatingHours	string	*optional* Operating Hours in h/yr input range: (0, 8760]
	Variable Speed Control Availability	VariableSpeedControlAvailability	string	*optional* Variable Speed Control Availability input range: {“Yes”, “No”}
	Air Leakage in Ducts	AirLeakageInDucts	string	*optional* Air Leakage in Ducts in % of airflow input range: [0, 100]
	Energy Consumption (Measured)	EnergyConsumptionMeasured	string	*optional* Energy Consumption (Measured) in kWh/yr input range: (0, ∞)
Lighting Systems	Lighting Power Density	LightingPowerDensity	string	*optional* Lighting Power Density in W/m² input range: (0, ∞)
	Operating Hours	OperatingHours	string	*optional* Operating Hours in h/yr input range: (0, 8760]

	Control Systems	ControlSystems	string	*optional* Control Systems input range: {“None”, “Timers”, “Occupancy Sensors”, “Daylight Harvesting”, “Combination”}
	Energy Consumption (Measured)	EnergyConsumptionMeasured	string	*optional* Energy Consumption (Measured) in kWh/yr input range: (0, ∞)
Water Heaters	Rated Capacity	RatedCapacity	string	*optional* Rated Capacity in kW input range: (0, ∞)
	Heating Efficiency	HeatingEfficiency	string	*optional* Heating Efficiency in % input range: [0, 100]
	Standby Heat Loss Rate	StandbyHeatLossRate	string	*optional* Standby Heat Loss Rate in % of stored energy per day input range: [0, 100]
	Scale Buildup Condition	ScaleBuildupCondition	string	*optional* Scale Buildup Condition input range: {“None”, “Light”, “Moderate”, “Heavy”}
	Operating Hours	OperatingHours	string	*optional* Operating Hours in h/yr input range: (0, 8760]
	Energy Consumption (Measured)	EnergyConsumptionMeasured	string	*optional* Energy Consumption (Measured) in kWh/yr input range: (0, ∞)
Chillers	Cooling Capacity	CoolingCapacity	string	*optional* Cooling Capacity in TR input range: (0, ∞)
	kW/TR Ratio	KWTRRatio	string	*optional* kW/TR Ratio in kW/TR input range: (0, ∞)
	Operating Hours	OperatingHours	string	*optional* Operating Hours in h/yr input range: (0, 8760]
	Condenser Condition	CondenserCondition	string	*optional* Condenser Condition input range: {“Clean”, “Fouled”, “Poor”}
	Chilled Water Temperature Difference	ChilledWaterTemperatureDifference	string	*optional* Chilled Water Temperature Difference in °C input range: (0, ∞)
	Energy Consumption (Measured)	EnergyConsumptionMeasured	string	*optional* Energy Consumption (Measured) in kWh/yr input range: (0, ∞)
Cooling Towers	Heat Rejection Capacity	HeatRejectionCapacity	string	*optional* Heat Rejection Capacity (see Heat Rejection Capacity Unit for the unit used) input range: (0, ∞)

	Heat Rejection Capacity Unit	HeatRejectionCapacityUnit	string	*optional* Heat Rejection Capacity Unit input range: {“kW”, “TR”}
	Fan Motor Efficiency	FanMotorEfficiency	string	*optional* Fan Motor Efficiency in % input range: [0, 100]
	Operating Hours	OperatingHours	string	*optional* Operating Hours in h/yr input range: (0, 8760]
	Approach Temperature	ApproachTemperature	string	*optional* Approach Temperature in °C input range: (0, ∞)
	VFD Availability	VFDAvailability	string	*optional* VFD Availability input range: {“Yes”, “No”}
	Cycles of Concentration	CyclesOfConcentration	string	*optional* Cycles of Concentration (ratio of dissolved solids in circulating water to make-up water) input range: (1, ∞)
	Energy Consumption (Measured)	EnergyConsumptionMeasured	string	*optional* Energy Consumption (Measured) in kWh/yr input range: (0, ∞)
Computers / Data Centers	IT Equipment Load	ITEquipmentLoad	string	*optional* IT Equipment Load in kW input range: (0, ∞)
	Power Usage Effectiveness	PowerUsageEffectiveness	string	*optional* Power Usage Effectiveness in PUE input range: [1, ∞)
	Server Utilization	ServerUtilization	string	*optional* Server Utilization in % input range: [0, 100]
	UPS Efficiency	UPSEfficiency	string	*optional* UPS Efficiency in % input range: [0, 100]
	Hot/Cold Aisle Management	HotColdAisleManagement	string	*optional* Hot/Cold Aisle Management input range: {“Yes”, “No”, “Partial”}
	Energy Consumption (Measured)	EnergyConsumptionMeasured	string	*optional* Energy Consumption (Measured) in kWh/yr input range: (0, ∞)
Elevators / Mechanical Equipment	Motor Efficiency	MotorEfficiency	string	*optional* Motor Efficiency in % input range: [0, 100]
	Drive Technology	DriveTechnology	string	*optional* Drive Technology input range: {“Geared”, “Hydraulic”, “Gearless”, “VFD-driven”}

	Regenerative Braking Availability	RegenerativeBrakingAvailability	string	*optional* Regenerative Braking Availability input range: {“Yes”, “No”}
	Operating Cycles per Day	OperatingCyclesPerDay	string	*optional* Operating Cycles per Day in cycles/day input range: (0, ∞)
	Energy Consumption per Cycle	EnergyConsumptionPerCycle	string	*optional* Energy Consumption per Cycle in kWh/cycle input range: (0, ∞)
	Energy Consumption (Measured)	EnergyConsumptionMeasured	string	*optional* Energy Consumption (Measured) in kWh/yr input range: (0, ∞)
Clinker Handling / Conveyors	Motor Power Rating	MotorPowerRating	string	*optional* Motor Power Rating in kW input range: (0, ∞)
	Motor Efficiency Class	MotorEfficiencyClass	string	*optional* Motor Efficiency Class input range: {“IE1”, “IE2”, “IE3”, “IE4”}
	Loading Factor	LoadingFactor	string	*optional* Loading Factor in % input range: [0, 100]
	Operating Hours	OperatingHours	string	*optional* Operating Hours in h/yr input range: (0, 8760]
	Variable Speed Drive Availability	VariableSpeedDriveAvailability	string	*optional* Variable Speed Drive Availability input range: {“Yes”, “No”}
	Idle Running Time	IdleRunningTime	string	*optional* Idle Running Time in % of operating time input range: [0, 100]
	Energy Consumption (Measured)	EnergyConsumptionMeasured	string	*optional* Energy Consumption (Measured) in kWh/yr input range: (0, ∞)
Pumps	Rated Power	RatedPower	string	*optional* Rated Power in kW input range: (0, ∞)
	Pump Efficiency	PumpEfficiency	string	*optional* Pump Efficiency in % input range: [0, 100]
	Operating Hours	OperatingHours	string	*optional* Operating Hours in h/yr input range: (0, 8760]
	Flow Control Method	FlowControlMethod	string	*optional* Flow Control Method input range: {“Throttling Valve”, “VFD”, “On-Off Control”, “Bypass/Recirculation”, “None”}
	Load Factor	LoadFactor	string	*optional* Load Factor in % input range: [0, 100]

	Energy Consumption (Measured)	EnergyConsumptionMeasured	string	*optional* Energy Consumption (Measured) in kWh/yr input range: (0, ∞)
Renewable Generation (Solar PV/Thermal /Wind)	Generation Type	GenerationType	string	*optional* Generation Type input range: {"Solar PV", "Solar Thermal", "Wind", "Other"}
	Installed Capacity	InstalledCapacity	string	*optional* Installed Capacity in kWp (Solar PV/Wind) or kWth (Solar Thermal) input range: (0, ∞)
	Performance Ratio	PerformanceRatio	string	*optional* Performance Ratio in % (actual output as a share of theoretical output; captures soiling, shading, degradation, and blade/panel wear) input range: [0, 100]
	Annual Generation (Measured)	AnnualGenerationMeasured	string	*optional* Annual Generation (Measured) in kWh/yr (Solar PV/Wind) or GJ/yr (Solar Thermal) input range: (0, ∞)
Water/Wastewater Treatment	Treatment Capacity	TreatmentCapacity	string	*optional* Treatment Capacity in m^3/day input range: (0, ∞)
	Annual Volume Treated	AnnualVolumeTreated	string	*optional* Annual Volume Treated in m^3/yr input range: (0, ∞)
	Specific Energy Consumption	SpecificEnergyConsumption	string	*optional* Specific Energy Consumption in kWh/m^3 treated input range: (0, ∞)
	Aeration System Type	AerationSystemType	string	*optional* Aeration System Type input range: {"Diffused Aeration", "Mechanical Surface Aerator", "None/Not Applicable"}
	Aeration Control	AerationControl	string	*optional* Aeration Control input range: {"Fixed Speed", "DO-based Automatic Control (VFD)", "None/Not Applicable"}
	Operating Hours	OperatingHours	string	*optional* Operating Hours in h/yr input range: (0, 8760]
	Energy Consumption (Measured)	EnergyConsumptionMeasured	string	*optional* Energy Consumption (Measured) in kWh/yr input range: (0, ∞)
Construction Site Energy	Diesel Consumption Intensity	DieselConsumptionIntensity	string	*optional* Diesel Consumption Intensity in L/m^2 GFA input range: (0, ∞)
	Grid Electricity Consumption Intensity	GridElectricityConsumptionIntensity	string	*optional* Grid Electricity Consumption Intensity in kWh/m^2 GFA input range: (0, ∞)
	Renewable Energy Share of Total Energy	RenewableEnergyShareOfTotalEnergy	string	*optional* Renewable Energy Share of Total Energy

				Consumption in % input range: [0, 100]
	Generator Load Factor	GeneratorLoadFactor	string	*optional* Average Diesel Generator Load Factor in % input range: [0, 100]

Response

Variable Name	Type	Description
errcode	int	Error code for the API (0: ok, 1: error)
errmsg	string	Message
TotalEnergyConsumption	string	Total energy consumption in GJ/year.
TotalCarbonEmission	string	Total carbon emission in t CO2e/year.
SpecificEnergyConsumptionBenchmarking	object	The Specific Energy Consumption (SEC), Best Available Techniques (BAT) benchmark, and benchmarking result (expressed as “On Track”, i.e., equal and better than BAT, or “Needs Improvement”, i.e., worse than BAT). SEC and SECOfBAT share the same unit as the industry's primary energy KPI (see SECUnit). For example, “SEC”: “4”, “SECOfBAT”: “3.4”, “SECUnit”: “GJ/t clinker”, “SECBenchmarkingResult”: “Needs Improvement”
KPIBenchmarking	object	The key point indicators specific to the IndustryType selected in the request, each with its BAT benchmark and benchmarking result. Each metric uses the same Variable Name as its corresponding “Facility KPIs - <Industry Type>” input parameter above (see there for the unit of each metric), paired with “<Name>OfBAT”, “<Name>BenchmarkingResult” and “<Name>Recommendation” keys; “<Name>Recommendation” is always present, shown as “-” when that KPI’s BenchmarkingResult is “On Track”. The specific KPI fields returned depend on IndustryType. For example, for IndustryType “Iron & Steel”: “ScrapRecoveryRate”: “90”, “ScrapRecoveryRateOfBAT”: “95”, “ScrapRecoveryRateBenchmarkingResult”: “Needs Improvement”, “ScrapRecoveryRateRecommendation”: [“Improve scrap sorting/grading and

		magnetic separation to reduce yield losses”]

SystemPerformanceBenchmarking	object	The efficiencies of the input systems, BAT benchmark and benchmarking result, grouped by system type. Each system is keyed by its Variable Name (matching the system objects above, e.g. “Furnaces”), and each metric inside uses the same Variable Name as the corresponding system input field (see there for the unit), paired with “<Name>OfBAT” and a shared “BenchmarkingResult” key. Only the systems selected in SystemTypes in the request are included. For example, for SystemTypes including “Furnaces”:
		
		“Furnaces”: {
		“ThermalEfficiency”: “82”,
		“ThermalEfficiencyOfBAT”: “85”,
		“BenchmarkingResult”: “Needs Improvement”
		}

EnergyAnalysisAndRecommendation	object	Estimated system energy consumption, recommended green measures and estimated energy savings, grouped by system type, plus an overall total. Each system is keyed by its Variable Name (e.g. “Furnaces”), containing “EnergyConsumption” and “EnergySavings” (both in GJ) and a “Recommendation” list; “TotalEnergySavings” (GJ) sits alongside the per-system objects. Only the systems selected in SystemTypes in the request are included. For example, for SystemTypes including “Furnaces”: “Furnaces”: { “EnergyConsumption”: “1116290”, “Recommendation”: [“Optimise combustion air control to reduce excess air ratio”, “Install flue gas waste heat recovery system”], “EnergySavings”: “167000” }, “TotalEnergySavings”: “171940” *Note: For “Renewable Generation (Solar PV/Thermal/Wind)”, “EnergyConsumption” is reported as a negative value (GJ) representing the estimated annual generation (offset). “EnergySavings” remains non-negative. Both Renewable Generation and Construction Site Energy (a*

		one-time total, not an annual value) are excluded from "TotalEnergySavings".
WaterAnalysisAndRecommendation	object	Water consumption (m^3), recommended green measures ("WaterRecommendation") and estimated water savings ("TotalWaterSavings", m^3), where applicable. For example, "WaterConsumption": "100000", "WaterRecommendation": ["Recover condensate"], "TotalWaterSavings": "20000"
RawMaterialAnalysisAndRecommendation	object	Raw material consumption (ton), recommended green measures ("RawMaterialRecommendation") and estimated raw material savings ("TotalRawMaterialSavings", ton), where applicable. For example, "RawMaterialConsumption": "1000", "RawMaterialRecommendation": ["Continuous casting optimization"], "TotalRawMaterialSavings": "50"

Examples

Example Request


```
{

    "ProjectName": "AAA",
    "ProjectAddress": "BBB",
    "ProjectPostalCode": "000000",
    "ProjectCountry": "Egypt",
    "ProjectGFA": "45000",
    "ProjectStatus": "Existing Building",
    "BuildingData":
        {
            "BuildingInfo":
                {
                    "BuildingName": "CCC",
                    "GFA": "45000",
                    "BuildingType": "Industrial Facility",
                    "IndustryType": "Cement",
                    "MonthlyProduction": "38000",
                    "Consumption":
                        {
                            "MonthlyElectricityConsumption": "1250000",
                            "MonthlyNaturalGasConsumption": "0",
                            "MonthlyCoalConsumption": "3200",
                            "MonthlyDieselConsumption": "15",
                            "MonthlyPetrolConsumption": "0",
                            "MonthlyLPGConsumption": "0",
                            "MonthlyLNGConsumption": "0",
                            "MonthlyHFOConsumption": "0",
                            "MonthlyLFOConsumption": "0",
                            "MonthlyOtherConsumption": "0",
                            "OtherEnergyEF": "0",
```

```
                        "MonthlyWaterConsumption": "8500",
                        "MonthlyRawMaterialConsumption": "42000"
                    }
                ,
                "FacilityKPIs":
                    {
                        "SpecificThermalEnergyConsumption": "3.6",
                        "SpecificElectricalEnergyConsumption": "95",
                        "CO2Intensity": "540",
                        "ClinkerRatio": "0.68",
                        "AlternativeFuelRate": "22"
                    }
            }
        ,
        "SystemTypes": ["Motors", "Furnaces", "Clinker Handling / Conveyors"],
        "Motors":
            {
                "RatedPower": "750",
                "MotorEfficiency": "88",
                "OperatingHours": "7200",
                "LoadFactor": "65",
                "VFDAvailability": "No",
                "EnergyConsumptionMeasured": "3985000"
            }
        ,
        "Furnaces":
            {
                "RatedCapacity": "45",
                "LoadFactor": "85",
                "OperatingHours": "8000",
                "ThermalEfficiency": "82",
                "ExcessAirRatio": "1.45",
                "FlueGasLosses": "18",
                "InsulationSurfaceTemperature": "68",
                "EnergyConsumptionMeasured": "309500000"
            }
        ,
        "ClinkerHandlingConveyors":
            {
                "MotorPowerRating": "160",
                "MotorEfficiencyClass": "IE2",
                "LoadingFactor": "55",
                "OperatingHours": "8000",
                "VariableSpeedDriveAvailability": "No",
                "IdleRunningTime": "28",
                "EnergyConsumptionMeasured": "3850000"
            }
    }
}
```


Example Response


```
{
    "result":
        {
```

```
        "errcode": 0,
        "errmsg": ""
    }
,
"data":
    {
        "TotalEnergyConsumption": "1051164",
        "TotalCarbonEmission": "102656.94",
        "SpecificEnergyConsumptionBenchmarking":
            {
                "SEC": "3.6",
                "SECOfBAT": "3.4",
                "SECUnit": "GJ/t clinker",
                "SECBenchmarkingResult": "Needs Improvement"
            }
        ,
        "KPIBenchmarking":
            {
                "SpecificThermalEnergyConsumption": "3.6",
                "SpecificThermalEnergyConsumptionOfBAT": "3.4",
                "SpecificThermalEnergyConsumptionBenchmarkingResult": "Needs
                Improvement",
                “SpecificThermalEnergyConsumptionRecommendation”: [“Upgrade to a 6-stage
                preheater/precalciner cyclone tower”],
                "SpecificElectricalEnergyConsumption": "95",
                "SpecificElectricalEnergyConsumptionOfBAT": "90",
                "SpecificElectricalEnergyConsumptionBenchmarkingResult": "Needs
                Improvement",
                “SpecificElectricalEnergyConsumptionRecommendation”: [“Replace ball mills
                with Vertical Roller Mills (VRM) or High-Pressure Grinding Rolls (HPGR) for finish
                grinding”],
                "CO2Intensity": "540",
                "CO2IntensityOfBAT": "520",
                "CO2IntensityBenchmarkingResult": "Needs Improvement",
                “CO2IntensityRecommendation”: [“Increase alternative/waste-derived fuel co-
                firing to displace fossil coal/petcoke”],
                "ClinkerRatio": "0.68",
                "ClinkerRatioOfBAT": "0.65",
                "ClinkerRatioBenchmarkingResult": "Needs Improvement",
                “ClinkerRatioRecommendation”: [“Increase fly ash, GGBS and limestone-filler
                substitution in cement blends”],
                "AlternativeFuelRate": "22",
                "AlternativeFuelRateOfBAT": "30",
                "AlternativeFuelRateBenchmarkingResult": "Needs Improvement",
                “AlternativeFuelRateRecommendation”: [“Develop an alternative-fuel (AFR)
                programme using tyres, biomass and RDF/SRF”, “Obtain the EEAA permits
                required for co-processing alternative fuels”]
            }
        ,
        "SystemPerformanceBenchmarking":
            {
                "Motors":
                    {
                        "MotorEfficiency": "88",
```

```
                    "MotorEfficiencyOfBAT": "90",
                    "BenchmarkingResult": "Needs Improvement"
                }
            ,
            "Furnaces":
                {
                    "ThermalEfficiency": "82",
                    "ThermalEfficiencyOfBAT": "85",
                    "BenchmarkingResult": "Needs Improvement"
                }
            ,
            "ClinkerHandlingConveyors":
                {
                    "MotorEfficiencyClass": "IE2",
                    "MotorEfficiencyClassOfBAT": "IE3",
                    "BenchmarkingResult": "Needs Improvement"
                }
        }
    ,
    "EnergyAnalysisAndRecommendation":
        {
            "Motors":
                {
                    "EnergyConsumption": "14360",
                    "Recommendation": ["Replace with IE3/IE4 premium-efficiency motor",
                    "Install VFD (load factor below 70%)"],
                    "EnergySavings": "2150"
                }
            ,
            "Furnaces":
                {
                    "EnergyConsumption": "1116290",
                    "Recommendation": ["Optimise combustion air control to reduce excess
                    air ratio", "Install flue gas waste heat recovery system", "Upgrade
                    refractory/insulation"],
                    "EnergySavings": "167000"
                }
            ,
            "ClinkerHandlingConveyors":
                {
                    "EnergyConsumption": "13950",
                    "Recommendation": ["Retrofit variable speed drive (VSD)", "Add
                    automated start/stop control to reduce idle running time", "Upgrade to
                    premium-efficiency (IE3) motor"],
                    "EnergySavings": "2790"
                }
            ,
            "TotalEnergySavings": "171940"
        }
    ,
    "WaterAnalysisAndRecommendation":
        {
            "WaterConsumption": "102000",
```

```
                "WaterRecommendation": ["Recover cooling tower blowdown and process
                water for reuse"],
                "TotalWaterSavings": "15000"
            }
        ,
        "RawMaterialAnalysisAndRecommendation":
            {
                "RawMaterialConsumption": "504000",
                "RawMaterialRecommendation": ["Improve raw meal grinding/blending
                efficiency to reduce material losses", "Increase fly ash, GGBS and limestone-filler
                substitution in cement blends"],
                "TotalRawMaterialSavings": "10000"
            }
    }
}
```